Online conformal inference for multi-step time series forecasting

Xiaoqian Wang*
Academy of Mathematics and Systems Science, Chinese Academy of Sciences
and
Rob J Hyndman
Department of Econometrics & Business Statistics, Monash University

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Abstract

We study the construction of distribution-free prediction intervals for multi-step time series forecasting. In this setting, forecast errors at different horizons are serially dependent, and feedback is delayed, since the realized error of an h-step-ahead forecast becomes available only after h periods. We propose autocorrelated multi-step conformal prediction (AcMCP), an online procedure that accommodates delayed feedback and explicitly incorporates the dependence among multi-step forecast errors into conformal calibration. To characterize this dependence, we derive first-order representations of the optimal forecast errors under a general nonstationary autoregressive data-generating process, together with explicit bounds on the remainder terms. AcMCP guarantees long-run coverage of at least the nominal level at every horizon, without requiring exchangeability or any assumption on how the data distribution evolves over time. We also establish a finite-sample bound on the empirical coverage shortfall, which depends explicitly on the forecast horizon and the evaluation sample size. Notably, these guarantees do not rely on the accuracy of the linear approximations. Simulations and real-data applications show that AcMCP adapts to time-varying dynamics and maintains rolling coverage close to the nominal level. By exploiting the dependence among multi-step errors, it often yields narrower intervals than its counterpart without this dependence, with similar coverage, particularly at longer horizons.

Keywords: Conformal prediction, Coverage guarantee, Distribution-free inference, Exchangeability, Delayed feedback

*Corresponding author. Xiaoqian Wang, Academy of Mathematics and Systems Science, Chinese Academy of Sciences, Beijing, 100190, China. E-mail address: xiaoqian.wang@amss.ac.cn (X. Wang).

1 Introduction

Conformal prediction (Papadopoulos et al. 2002, Vovk et al. 2005) is a model-agnostic framework for uncertainty quantification that constructs prediction intervals with valid coverage, without requiring stringent assumptions on the underlying data distribution. Its classical guarantees, however, rely on the data being i.i.d. or, more generally, exchangeable. In time series forecasting, observations are typically serially dependent and may exhibit nonstationarity or distributional shifts, so exchangeability generally fails and the standard guarantees no longer hold.

Recent work has sought to relax this requirement. One line of research replaces exchangeability with weaker, time-series-specific assumptions. To handle temporal dependence, approximate coverage guarantees have been obtained through permutation schemes that preserve the temporal dependence structure (Chernozhukov et al. 2018, Yu et al. 2022) and through concentration and decoupling arguments for split conformal prediction (Oliveira et al. 2024). Xu & Xie (2021) use bootstrap ensemble predictors to obtain out-of-bag residuals and update their empirical distribution as new observations arrive, establishing approximate coverage under stationary, strongly mixing errors. Under covariate shift, Tibshirani et al. (2019) and Yang, Kuchibhotla & Tchetgen Tchetgen (2024) establish coverage guarantees when the training and test distributions differ only in their covariate marginals. Other work generalizes conformal calibration by replacing the unweighted empirical distribution of nonconformity scores with a weighted one (Barber et al. 2023, Guan 2023).

A separate strand of research develops online conformal methods that adjust prediction intervals using sequential coverage feedback. Adaptive conformal inference (ACI, Gibbs & Candès 2021) updates the working miscoverage level in response to observed coverage errors, so that empirical coverage converges to the target level without distributional assumptions on the time series. Subsequent extensions use expert aggregation and adaptive online learning to reduce sensitivity to the learning rate and to improve adaptation to distribution shifts (Zaffran et al. 2022, Gibbs &

Candès 2024, Bhatnagar et al. 2023). Conformal PID control (Angelopoulos et al. 2023) builds on the same feedback principle but adjusts the thresholds of the nonconformity scores directly, combining quantile tracking, correction for cumulative coverage errors, and a scorecaster that exploits predictable temporal structure in the scores. Further refinements improve stability through decaying step sizes (Angelopoulos et al. 2024), reduce tuning through parameter-free updates (Podkopaev et al. 2024), and incorporate smooth feedback that reflects the magnitude of the discrepancy between scores and thresholds (Wu et al. 2025).

Most of this literature, however, targets single-step forecasting, whereas many applications require uncertainty quantification over several future horizons. Multi-step forecasting poses two additional challenges for conformal methods. First, feedback is delayed: the realized error of an h-step-ahead forecast becomes available only h periods after the forecast is issued, so online updates must rely on outdated information. Second, forecast errors at different horizons issued from the same origin are dependent, because errors propagate through the forecasting recursion.

Existing conformal methods for multi-step forecasting address these challenges only partially. Stankeviciute et al. (2021) integrate conformal prediction with recurrent neural networks for multi-step forecasting and apply a Bonferroni correction to obtain simultaneous coverage across horizons, but their guarantee relies on exchangeability across time series. Schlembach et al. (2025) extend this approach to multivariate targets under nonexchangeability, with a joint miscoverage bound across horizons and response dimensions. Bellman conformal inference (Yang, Candès & Lei 2024) uses multi-step forecasts in a look-ahead optimization that trades off interval length against a miscoverage penalty, but it outputs, and guarantees coverage for, only the resulting sequence of one-step-ahead intervals; it also requires nested interval families, which calls for additional construction when only point forecasts are available, and solves a dynamic program at every time step. The multi-step extension of sequential predictive conformal inference (SPCI, Xu & Xie 2023) fits a separate predictor and conditional quantile model for each horizon and uses past residuals to adapt the interval bounds; although SPCI has asymptotic conditional coverage

guarantees under suitable assumptions, its multi-step implementation is not analyzed separately. To our knowledge, existing multi-step conformal methods seldom provide coverage guarantees without exchangeability, and none exploits the dependence among forecast errors across horizons. A method that handles delayed feedback, accounts for this dependence, and retains distribution-free coverage guarantees is therefore still lacking.

We consider a time series $\{y_t\}_{t\geq 1}$ generated by an unknown, possibly nonstationary data-generating process (DGP) that may depend on its own past and on exogenous predictors $\boldsymbol{x}_t \in \mathbb{R}^p$. At each time t, we wish to construct a prediction interval $\hat{\mathscr{C}}_{t+h|t}$ for y_{t+h} at every horizon $h \in [H]$. Given a user-specified miscoverage level $\alpha \in (0,1)$, the goal is horizon-specific long-run coverage, namely $\liminf_{T\to\infty} \frac{1}{T}\sum_{t=1}^{T} \mathbb{1}\left\{y_{t+h} \in \hat{\mathscr{C}}_{t+h|t}\right\} \geq 1-\alpha$, for $h \in [H]$.

This paper makes three contributions. First, we develop multi-step conformal prediction methods within an online learning framework with sequential training and calibration splits. Building on conformal PID control of Angelopoulos et al. (2023), we introduce its multi-step extension (MPID), which accommodates delayed coverage feedback. Second, we propose autocorrelated multi-step conformal prediction (AcMCP), which explicitly incorporates the dependence among multi-step forecast errors into the scorecaster. To motivate its design, we analyze optimal forecast errors under a general nonstationary autoregressive DGP with future exogenous predictors treated as known, derive first-order error decompositions, and bound the Taylor remainders under suitable smoothness and moment conditions. These results lead to a scorecaster that combines a moving average model for errors at a fixed horizon with a regression on errors across horizons from the same forecast origin. AcMCP combines these error forecasts with online calibration of the two tails to construct asymmetric prediction intervals for each horizon. Third, for both MPID and AcMCP, we establish finite-sample bounds on empirical coverage that explicitly account for delayed feedback and imply long-run coverage of at least the nominal level at each fixed horizon. These guarantees hold under suitable conditions on the scores and the calibration procedure, without any assumption on how the data distribution changes over time. They concern

coverage averaged over time at each horizon, rather than simultaneous coverage across horizons or conditional coverage at individual forecast origins. Importantly, the coverage guarantees of AcMCP do not depend on the accuracy of the forecast-error approximations; the contribution of the scorecaster to interval efficiency is assessed empirically. Simulations and applications to electricity demand and eating-out expenditure show that AcMCP adapts to distributional changes and often produces narrower intervals than MPID with similar coverage, particularly at longer horizons. A post-hoc comparison shows that this advantage does not extend uniformly to an adaptive benchmark whose step size is tuned on the test set.

We developed the `conformalForecast` package for R to implement the proposed multi-step conformal prediction methods. The package is publicly available on CRAN. All the data and code to reproduce the experiments are made available at https://github.com/xqnwang/cpts.

2 Online learning with sequential splits

Problem setup. We observe a time series $\{y_t\}_{t\geq 1}$ generated by an unknown DGP, which may depend on its own past as well as on exogenous predictors $\boldsymbol{x}_t = (x_{1,t},\dots,x_{p,t})' \in \mathbb{R}^p$ and their histories. Write $z_t = (\boldsymbol{x}_t, y_t)$. The joint distribution of $\{z_t\}_{t\geq 1}$ is allowed to vary over time, thereby accommodating nonstationary processes, and we impose no assumption on how it evolves. At each time t, we forecast H steps ahead and, for each $h \in [H]$, construct a *prediction set* $\hat{\mathscr{C}}^{\alpha}_{t+h|t}$ for y_{t+h}, which is a prediction interval in this setting. The h-step-ahead forecast uses the observed data $\{z_i\}_{1\leq i\leq t}$ together with the exogenous predictors $\{\boldsymbol{x}_{t+j}\}_{1\leq j\leq h}$ over the forecast period. These future predictors may be replaced by forecasts based on information available up to and including time t (*ex-ante* forecasts), or the actual values may be assumed available (*ex-post* forecasts). Given a user-specified *miscoverage level* $\alpha \in (0,1)$, we seek intervals with horizon-specific long-run coverage,

$$\liminf_{T\to\infty} \frac{1}{T-h} \sum_{t=h+1}^{T} \mathbb{1}\left\{y_t \in \hat{\mathscr{C}}^{\alpha}_{t|t-h}\right\} \geq 1-\alpha, \qquad h \in [H]. \tag{1}$$

This criterion concerns coverage averaged over time at each forecast horizon, rather than coverage conditional on the available information or simultaneous coverage of the entire future path. The parametric form of the DGP is assumed only for the forecast-error analysis in Section 4.1; the coverage guarantees in Section 4.3 do not require it.

Forecasts and errors. Suppose that, at each time t, a forecasting model $\hat{f}_t$ is trained on historical data $z_{1:t}$ and updated (retrained) as new observations arrive. Using $\hat{f}_t$, we produce point forecasts $\hat{y}_{t+h|t}$ and signed forecast errors $e_{t+h|t} = y_{t+h} - \hat{y}_{t+h|t}$ for $h \in [H]$. Conformal inference then calibrates these errors to turn the point forecasts into prediction intervals.

Prediction intervals. We construct generally asymmetric intervals by calibrating the lower and upper tails separately. Choose tail levels $\alpha^-, \alpha^+ > 0$ satisfying $\alpha^- + \alpha^+ = \alpha$. The two calibration procedures use the *nonconformity scores* $s^-_{t+h|t} = -e_{t+h|t}$ and $s^+_{t+h|t} = e_{t+h|t}$, respectively, and produce thresholds $q^-_{t+h|t}$ and $q^+_{t+h|t}$. These thresholds define the prediction interval

$$\hat{\mathscr{C}}^{\alpha}_{t+h|t} := \left[\hat{y}_{t+h|t} - q^-_{t+h|t},\ \hat{y}_{t+h|t} + q^+_{t+h|t}\right]. \tag{2}$$

For each tail $a \in \{-,+\}$, the *miscoverage indicator* $\mathrm{err}^a_{t+h|t} := \mathbb{1}\{s^a_{t+h|t} > q^a_{t+h|t}\}$ records whether the observation falls beyond the corresponding interval bound. Using opposite signed scores allows both tails to be calibrated through the same score-exceedance rule, with observations at either endpoint counted as covered. Because y_{t+h}, and hence $s^a_{t+h|t}$ and $\mathrm{err}^a_{t+h|t}$, is revealed only at time $t+h$, feedback is delayed: at time t, the information available for horizon h comes from forecast origins $t-h$ and earlier. We use $\alpha^- = \alpha^+ = \alpha/2$ throughout the empirical analysis.

Unless otherwise stated, we present the methods and theoretical results in subsequent sections for a generic single tail, omitting the tail superscript for simplicity. Accordingly, α denotes the target tail miscoverage level in single-tail statements and the total miscoverage level in two-sided statements. For a candidate response y, $s^y_{t+h|t}$ denotes the score obtained by replacing y_{t+h} with y.

Sequential split. In a time series context, it is inappropriate to perform *random splitting*, a standard strategy in much of the conformal prediction literature, due to the temporal dependency

present in the data. Following related work such as Wisniewski et al. (2020) and Zaffran et al. (2022), all conformal methods proposed in this paper use a *sequential split* to preserve the temporal structure. For example, the t available data points, $z_{1:t}$, are sequentially split into two consecutive sets, a *proper training set* $\mathscr{D}_{\text{tr}} \subset \{1,\dots,t_r\}$ and a *calibration set* $\mathscr{D}_{\text{cal}} \subset \{t_r+1,\dots,t\}$, where $t_c = t - t_r \gg H$, $\mathscr{D}_{\text{tr}} \cup \mathscr{D}_{\text{cal}} = \{1,\dots,t\}$, and $\mathscr{D}_{\text{tr}} \cap \mathscr{D}_{\text{cal}} = \varnothing$. Here, "proper" means that the training set is used exclusively for fitting the model, with no overlap with the calibration set, which is essential for the validity of coverage in conformal prediction (Papadopoulos et al. 2002, Vovk et al. 2005). With sequential splitting, multiple H-step forecasts and their nonconformity scores can be computed on the calibration set.

Online learning. We adopt the following generic online learning framework, closely related to Zaffran et al. (2022), for all conformal methods discussed in later sections. The procedure is illustrated in Figure 1. This framework updates prediction intervals as new data points arrive, allowing us to assess their long-run coverage behavior. It adopts a standard rolling window evaluation strategy, although expanding windows could easily be used instead.

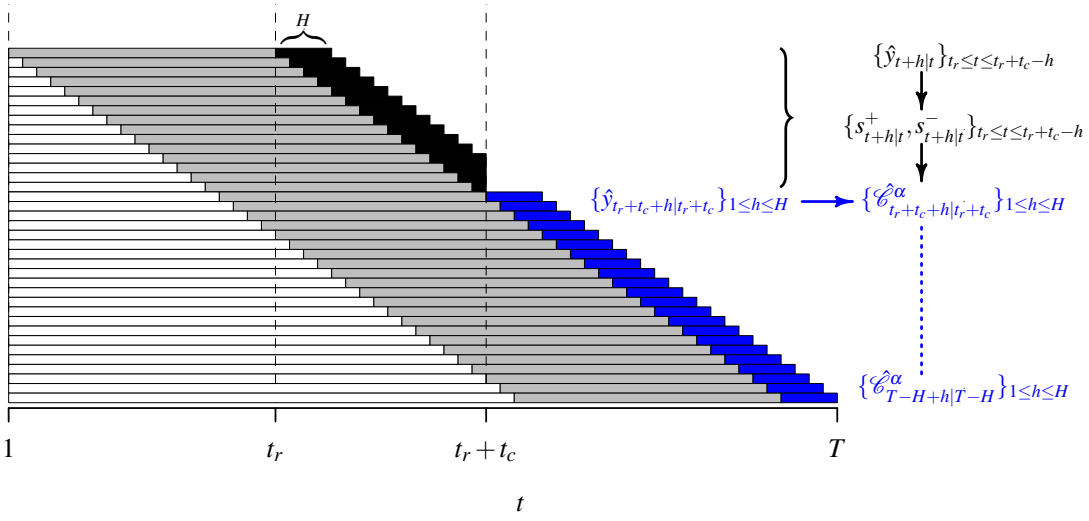


Figure 1: *Online learning framework with sequential splits. White: unused data; Gray: training data; Black: forecasts in the calibration set; Blue: forecasts in the test set.*

1. *Initialization*. Train a forecasting model on the initial proper training set $z_{(1+t-t_r):t}$, setting $t = t_r$. Then generate H-step-ahead point forecasts $\{\hat{y}_{t+h|t}\}_{h\in[H]}$ and compute the corre-

sponding upper- and lower-tail nonconformity scores $\{s^{+}_{t+h|t}, s^{-}_{t+h|t}\}_{h\in[H]}$ based on the true values $\{y_{t+h}\}_{h\in[H]}$.

2. *Recurring procedure*. Roll the training set forward by one data point by setting $t \to t+1$. Then repeat step 1 until the nonconformity scores on the entire initial calibration set, $\{s^{+}_{t+h|t}, s^{-}_{t+h|t}\}_{t_r\le t\le t_r+t_c-h}$ for $h \in [H]$, are computed.

3. *Quantile estimation and prediction interval calculation*. For each tail, use the nonconformity scores from the calibration set to estimate horizon-specific quantiles. Combine the two one-sided thresholds via Equation (2) to compute H-step-ahead prediction intervals on the test set.

4. *Online updating*. Continuously roll the training and calibration sets forward, one data point at a time, to update the nonconformity scores used for calibration, and repeat step 3 until prediction intervals for the entire test set are obtained. These intervals are $\{\hat{\mathscr{C}}^{\alpha}_{t+h|t}\}_{t_r+t_c\le t\le T-H}$ for $h \in [H]$, where $T - t_r - t_c$ is the length of the test set used to evaluate coverage against the long-run target in Equation (1).

3 Multi-step conformal PID control

For a well-specified forecasting model, the conditional mean minimizes the expected squared forecast error, and for covariance-stationary linear processes the variance of the resulting error $e_{t+h|t}$ is non-decreasing in h (Diebold & Lopez 1996, Patton & Timmermann 2007). The distribution of nonconformity scores therefore differs across horizons, so the thresholds should be calibrated separately for each $h \in [H]$. Calibration also need to cope with delayed feedback: at time t, the realized h-step score is available only for forecasts issued at origin $t-h$ or earlier.

We introduce the **multi-step conformal PID control** method (hereafter MPID), which extends the PID method (Angelopoulos et al. 2023), originally developed for one-step-ahead forecasting, to the multi-step setting within the online learning framework of Section 2. For each forecast

horizon $h \in [H]$, define the P-control state

$$q^{\mathrm{P}}_{t+h|t} = q^{\mathrm{P}}_{t+h-1|t-1} + \eta(\mathrm{err}_{t|t-h} - \alpha), \tag{3}$$

with a finite initial state $q^{\mathrm{P}}_{2h|h}$, typically zero. The estimated $1-\alpha$ quantile of the h-step-ahead score at time t is updated as

$$q_{t+h|t} = \underbrace{q^{\mathrm{P}}_{t+h|t}}_{\mathrm{P}} + \underbrace{r_t\left(\sum_{i=h+1}^{t}(\mathrm{err}_{i|i-h} - \alpha)\right)}_{\mathrm{I}} + \underbrace{\hat{s}_{t+h|t}}_{\mathrm{D}}, \tag{4}$$

where $\eta > 0$ is a learning rate, which may vary over time provided it is computed from information available at time t, and r_t is a saturation function that adheres to

$$x \geq c \cdot g(t-h) \Longrightarrow r_t(x) \geq b, \quad \text{and} \quad x \leq -c \cdot g(t-h) \Longrightarrow r_t(x) \leq -b, \tag{5}$$

for constants $b, c > 0$ and an admissible function g that is nonnegative, nondecreasing and sublinear, i.e., $g(n)/n \to 0$. For $h = 1$, Equation (4) reduces to the standard PID update. For each $h \in [H]$, we apply this update separately to the lower and upper tails, using the tail levels α^- and α^+ and the feedback available at time t, and combine the resulting thresholds via Equation (2) to form the prediction interval for y_{t+h}.

The three components play distinct roles.

P (quantile tracking). The P state moves up or down according to whether y_t fell outside or inside the one-sided prediction set issued at time $t-h$. It thus incorporates the h-period feedback delay explicitly: the most recent feedback usable at time t is $\mathrm{err}_{t|t-h}$.

I (error integration). The I term accumulates the historical coverage errors of the h-step-ahead one-sided prediction sets, during updates, thereby enhancing the stability of coverage.

D (scorecasting). The D term $\hat{s}_{t+h|t}$ is a forecast of the h-step-ahead nonconformity score, produced by any suitable forecasting model (a "scorecaster"; see Section 4) trained on the h-step-ahead scores available up to and including time t. Its effectiveness depends on the ability of the scorecaster to capture systematic and predictable structure in the scores, such as temporal

dependence or conditional heteroskedasticity. Only when such structure is present and adequately modeled does the D term provide additional benefits; otherwise, it can increase the variability of coverage and of the prediction sets.

MPID is a special case of the generic update in Equation (13), with $\hat{q}_{t+h|t} = q^{\mathrm{P}}_{t+h|t} + \hat{s}_{t+h|t}$. Its one- and two-sided prediction sets therefore inherit the finite-sample empirical coverage bounds and long-run guarantees of Corollary 2 and Corollary 3, respectively, under the stated assumptions. For reference, Appendix A1 presents multi-step extensions of split conformal prediction (Vovk et al. 2005), conformal prediction with weighted quantiles (Barber et al. 2023), and ACI (Gibbs & Candès 2021); we do not establish coverage guarantees for these.

4 Autocorrelated multi-step conformal prediction

MPID calibrates each horizon separately, and its scorecaster forecasts the scores of a single horizon. In any non-trivial context, however, multi-step forecast errors are correlated: the h-step-ahead error $e_{t+h|t}$ is correlated with the errors at the previous $h-1$ steps from the same origin, because errors propagate through the recursive forecasting procedure. No existing conformal method exploits this dependence.

This section studies the structure of multi-step forecast errors, uses it to design the AcMCP scorecaster, and establishes coverage guarantees that do not depend on how accurately the scorecaster approximates the errors.

4.1 Properties of multi-step forecast errors

We assume that a time series $\{y_t\}_{t\geq 1}$ follows a general nonstationary autoregressive process:

$$y_t = f_t(y_{(t-d):(t-1)}, \boldsymbol{x}_{(t-k):t}) + \varepsilon_t, \tag{6}$$

where f_t depends on d lagged values of y_t and on the current and k preceding values of the exogenous predictors, and ε_t is the innovation. To isolate uncertainty in future responses, we treat

future exogenous predictors as known. Let $\mathscr{F}_t$ be the information generated by the responses observed through time t and the exogenous predictor path. The optimal point forecast under squared loss is $\hat{y}_{t+h|t} = \mathbb{E}(y_{t+h}|\mathscr{F}_t)$, with error $e_{t+h|t} = y_{t+h} - \hat{y}_{t+h|t}$.

At origin t, $p_h = \min\{d, h-1\}$ of the lagged responses entering f_{t+h} are unobserved. For $h \geq 2$, collect them in $\boldsymbol{z}_{t,h} = (y_{t+h-1}, \ldots, y_{t+h-p_h})'$, with conditional mean $\boldsymbol{m}_{t,h} = \mathbb{E}(\boldsymbol{z}_{t,h}|\mathscr{F}_t)$, and let $F_{t,h}$ denote f_{t+h} as a function of $\boldsymbol{z}_{t,h}$ with all other arguments held fixed. We write ∂_i for differentiation in the ith coordinate, and $\|\cdot\|_2$ and $\|\cdot\|_{\text{op}}$ for the Euclidean and operator norms, respectively.

Proposition 1 (Autoregressive representation of optimal multi-step forecast errors)**.** *Consider the process in Equation* (6)*. Assume that each f_s is twice continuously differentiable in the lagged responses, that $\mathbb{E}(y_s^2) < \infty$ and $\mathbb{E}(\varepsilon_s|\mathscr{F}_{s-1}) = 0$ for all $s \geq 1$, and that the first-order Taylor terms and remainder below are integrable. Then, for each $h \in [H]$,*

$$e_{t+h|t} = \varepsilon_{t+h} + \sum_{i=1}^{p_h} \phi_{t,h,i} e_{t+h-i|t} + \widetilde{R}_{t,h}, \tag{7}$$

where $\phi_{t,h,i} = \partial_i F_{t,h}(\boldsymbol{m}_{t,h})$ and $\widetilde{R}_{t,h} = R_{t,h} - \mathbb{E}(R_{t,h}|\mathscr{F}_t)$, with $R_{t,h}$ the remainder of the first-order Taylor expansion of $F_{t,h}$ at $\boldsymbol{m}_{t,h}$. In particular, $\mathbb{E}(\widetilde{R}_{t,h}|\mathscr{F}_t) = 0$ and $\widetilde{R}_{t,1} = 0$.

Moreover, for $h \geq 2$, if there is a finite deterministic $M_{t,h} \geq 0$ with

$$\sup_{u\in[0,1]} \left\| \nabla^2 F_{t,h}\big(\boldsymbol{m}_{t,h} + u(\boldsymbol{z}_{t,h} - \boldsymbol{m}_{t,h})\big) \right\|_{\text{op}} \leq M_{t,h} \quad \textit{almost surely},$$

then

$$\mathbb{E}(|\widetilde{R}_{t,h}|\,|\,\mathscr{F}_t) \leq M_{t,h} \sum_{i=1}^{p_h} \mathbb{E}(e_{t+h-i|t}^2|\mathscr{F}_t). \tag{8}$$

Proof. See Appendix A2. □

It is well-established in the forecasting literature that, for a zero-mean covariance-stationary time series, the optimal linear least-squares forecasts have h-step-ahead errors that are at most MA$(h-1)$ process (Harvey et al. 1997), a property that can be derived using Wold's representation

theorem. For the general nonstationary autoregressive setting considered here, Proposition 2 gives an innovation representation with an explicit remainder.

Proposition 2 (Innovation representation of optimal multi-step forecast errors)**.** *Under the assumptions of Proposition 1, for each* $h \in [H]$,

$$e_{t+h|t} = \varepsilon_{t+h} + \sum_{i=1}^{h-1} \theta_{t,h,i}\varepsilon_{t+h-i} + U_{t,h}, \tag{9}$$

where the coefficients $\theta_{t,h,i}$ *are* $\mathscr{F}_t$*-measurable and the propagated remainders satisfy* $U_{t,1} = 0$ *and* $U_{t,j} = \widetilde{R}_{t,j} + \sum_{i=1}^{p_j} \phi_{t,j,i} U_{t,j-i}$ *for* $j \geq 2$.

Moreover, for $h \geq 2$*, suppose that the Hessian bounds of Proposition 1 hold and that, for finite deterministic constants* $M, L, V \geq 0$ *and all* $2 \leq j \leq h$,

$$M_{t,j} \leq M, \qquad \sum_{i=1}^{p_j} |\phi_{t,j,i}| \leq L, \qquad \sum_{i=1}^{p_j} \mathbb{E}(e^2_{t+j-i|t} | \mathscr{F}_t) \leq V \quad \textit{almost surely}.$$

Then

$$\mathbb{E}\big(|U_{t,h}| \,|\, \mathscr{F}_t\big) \leq MV \sum_{k=0}^{h-2} L^k \quad \textit{almost surely}. \tag{10}$$

Proof. See Appendix A3. □

Dropping the remainders in Propositions 1 and 2 gives linear approximations that motivate regression and moving-average specifications for multi-step errors, respectively. If the autoregressive functions are linear in their lagged responses, allowing intercepts, both remainders vanish and the representations are exact. Otherwise, Equation (8) controls the Taylor remainder through curvature and shorter-horizon mean squared errors, and Equation (10) controls its propagation across horizons.

Appendix A4 shows that, with t, h, and d fixed, both remainders have mean absolute size $O(\delta^2)$ and are therefore negligible relative to the innovation scale δ as $\delta \downarrow 0$. This requires deterministic bounds on gradients and Hessians along the expansion segments that do not depend on δ, and innovation conditional variances that are uniformly $O(\delta^2)$. The analysis motivates the AcMCP

scorecaster in Section 4.2, a simplified specification that accounts for dependence among forecast errors rather than reproducing their optimal dynamics exactly.

4.2 The AcMCP method

Motivated by these properties, we propose **autocorrelated multi-step conformal prediction** (AcMCP). Whereas multi-step extensions of existing conformal methods in Section 3 treat horizons separately, AcMCP exploits the dependence among multi-step forecast errors. For each $h \in [H]$, the h-step-ahead threshold is updated online as

$$q_{t+h|t} = q^{\mathrm{P}}_{t+h|t} + r_t\left(\sum_{i=h+1}^{t}(\mathrm{err}_{i|i-h} - \alpha)\right) + \tilde{s}_{t+h|t}, \tag{11}$$

where the lower- and upper-tail thresholds are updated separately at levels α^- and α^+ and combined into a prediction interval via Equation (2). AcMCP thus differs from MPID only in its scorecaster, which accounts for error dependence.

Let $\tilde{e}_{t+h|t}$ denote the forecast of $e_{t+h|t}$, obtained by equally combining two simple models: (i) an MA$(h-1)$ model fitted to the h-step-ahead errors available up to and including time t, namely $e_{1+h|1}, \ldots, e_{t|t-h}$, which captures serial dependence within a horizon; and (ii) a linear regression of each h-step-ahead error on the shorter-horizon errors from the same forecast origin, which captures dependence across horizons and takes the recursively predicted errors $\tilde{e}_{t+h-1|t}, \ldots, \tilde{e}_{t+1|t}$ as inputs, since the corresponding realized errors are not yet observed at time t. The score forecast in Equation (11) is $\tilde{s}_{t+h|t} = \tilde{e}_{t+h|t}$ for upper-tail calibration and $\tilde{s}_{t+h|t} = -\tilde{e}_{t+h|t}$ for lower-tail calibration. Importantly, the role of $\tilde{e}_{t+h|t}$ is not to forecast the errors precisely, but to incorporate the autocorrelation structure of multi-step forecast errors into the resulting prediction intervals.

The coverage guarantees below hold under the stated calibration conditions and do not require the linear approximations in Propositions 1 and 2 to be accurate.

4.3 Coverage guarantees

Proposition 3 and Corollaries 1, 2 concern single-tail calibration with fixed target tail miscoverage level α. Corollary 3 combines the lower and upper tails at levels α^- and α^+ with $\alpha = \alpha^- + \alpha^+$ the total miscoverage level.

Proposition 3. *Fix $h \in [H]$ and $\alpha \in (0,1)$. Let $\{s_{t+h|t}\}_{t\geq 1}$ be any sequence of finite scores in $(-b,b)$ with $b \in (0,\infty]$, and let r_t satisfy Equation (5) and an admissible function g. Then the iteration $q_{t+h|t} = r_t\left(\sum_{i=h+1}^{t}(\mathrm{err}_{i|i-h} - \alpha)\right)$ satisfies, for every $T \geq h+1$,*

$$\left|\frac{1}{T-h}\sum_{t=h+1}^{T}(\mathrm{err}_{t|t-h} - \alpha)\right| \leq \frac{c\cdot g(T-h)+h}{T-h}, \tag{12}$$

with c the constant in Equation (5). Consequently, $\lim_{T\to\infty}\frac{1}{T-h}\sum_{t=h+1}^{T}\mathrm{err}_{t|t-h} = \alpha$.

Proof. See Appendix A5. □

For $h = 1$, Proposition 3 recovers Proposition 2 of Angelopoulos et al. (2023). With $n = T-h$ evaluated forecasts, the bound decomposes into $c\,g(n)/n$, determined by the integrator's saturation, and h/n, which captures the feedback delay. For fixed n, a longer horizon enlarges the delay term, although actual calibration errors need not grow. For every fixed h the bound vanishes because g is sublinear, so the delay affects the finite-sample guarantee but not long-run calibration.

The constant-step quantile iteration is a special case of Proposition 3, so its calibration bound follows directly.

Corollary 1. *Fix $h \in [H]$ and $\alpha \in (0,1)$. Let $\{s_{t+h|t}\}_{t\geq 1}$ be any sequence of finite scores in $(-b,b)$ with $0 < b < \infty$, and let $q_{2h|h} \in \mathbb{R}$ be an arbitrary initial threshold. Then the iteration $q_{t+h|t} = q_{t+h-1|t-1} + \eta(\mathrm{err}_{t|t-h} - \alpha)$ with $\eta > 0$ satisfies, for every $T \geq h+1$,*

$$\left|\frac{1}{T-h}\sum_{t=h+1}^{T}(\mathrm{err}_{t|t-h} - \alpha)\right| \leq \frac{b + |q_{2h|h}| + \eta h}{\eta(T-h)}.$$

In particular, $\lim_{T\to\infty}\frac{1}{T-h}\sum_{t=h+1}^{T}\mathrm{err}_{t|t-h} = \alpha$.

Proof. See Appendix A6. □

For $h = 1$, Corollary 1 recovers Proposition 1 of Angelopoulos et al. (2023). The multi-step bound, $(b + |q_{2h|h}|)/(\eta n) + h/n$, separates the effects of score magnitude, initialization, and learning rate from that of the feedback delay. The boundedness of the scores is a purely theoretical assumption that keeps the accumulated miscoverage error of the quantile-tracking recursion under control. In practice, the bound itself need not be known, and the update runs unchanged as long as the scores are finite.

More importantly, Proposition 3 also establishes one-sided calibration for AcMCP. We first write Equation (11) as

$$q_{t+h|t} = \hat{q}_{t+h|t} + r_t \left(\sum_{i=h+1}^{t} (\mathrm{err}_{i|i-h} - \alpha) \right), \tag{13}$$

where $\hat{q}_{t+h|t}$ may be any function of the past observations $\{(x_i, y_i)\}_{1 \le i \le t}$ and the earlier thresholds $q_{i+h|i}$, $i \le t-1$, and can be viewed as a forecast of the next threshold from the available history. Taking $\hat{q}_{t+h|t} = q^{\mathrm{P}}_{t+h|t} + \tilde{s}_{t+h|t}$ recovers AcMCP, and taking $\hat{q}_{t+h|t} = q^{\mathrm{P}}_{t+h|t} + \hat{s}_{t+h|t}$ recovers MPID.

Corollary 2. *Fix $h \in [H]$ and $\alpha \in (0,1)$. Let the scores $s_{t+h|t}$ and baselines $\hat{q}_{t+h|t}$ be finite with $|s_{t+h|t} - \hat{q}_{t+h|t}| < b$ for all $t \ge 1$, where $b \in (0, \infty]$, and let r_t satisfy Equation (5) and an admissible function g. Then the iteration in Equation (13) satisfies, for every $T \ge h+1$,*

$$\left| \frac{1}{T-h} \sum_{t=h+1}^{T} (\mathrm{err}_{t|t-h} - \alpha) \right| \le \frac{c \cdot g(T-h) + h}{T-h}.$$

In particular, $\lim_{T\to\infty} \frac{1}{T-h} \sum_{t=h+1}^{T} \mathrm{err}_{t|t-h} = \alpha$.

Proof. See Appendix A7. □

Under the stated conditions, integral feedback ensures calibration for any baseline $\hat{q}_{t+h|t}$, and thus without requiring accurate forecasts from the scorecaster. The scorecaster instead aims to improve interval efficiency by accounting for error dependence, although this improvement is not guaranteed by the corollary. Unlike Corollary 1, which bounds the scores themselves, Corollary 2 bounds only their deviations from the baselines, $s_{t+h|t} - \hat{q}_{t+h|t}$.

The next corollary combines the lower- and upper-tail calibration guarantees to establish

coverage for the asymmetric prediction interval in Equation (2).

Corollary 3. *Fix $h \in [H]$ and $\alpha \in (0,1)$. Construct the interval in Equation (2) from thresholds obtained by applying Equation (13) separately to the lower- and upper-tail scores, with targets $\alpha^-, \alpha^+ > 0$ satisfying $\alpha^- + \alpha^+ = \alpha$. Suppose each tail satisfies the assumptions of Corollary 2. Then, for every $T \geq h+1$,*

$$\frac{1}{T-h} \sum_{t=h+1}^{T} \mathbb{1}\{y_t \in \hat{\mathscr{C}}^{\alpha}_{t|t-h}\} \geq 1 - \alpha - 2\frac{c \cdot g(T-h) + h}{T-h}, \tag{14}$$

and hence $\liminf_{T\to\infty} \frac{1}{T-h} \sum_{t=h+1}^{T} \mathbb{1}\{y_t \in \hat{\mathscr{C}}^{\alpha}_{t|t-h}\} \geq 1 - \alpha$.

Furthermore, let $\chi_{T,h} := \frac{1}{T-h} \sum_{t=h+1}^{T} \mathbb{1}\{\hat{y}_{t|t-h} - q^{-}_{t|t-h} > \hat{y}_{t|t-h} + q^{+}_{t|t-h}\}$ *be the frequency of crossed endpoints, that is, of empty intervals. Then*

$$\left| \frac{1}{T-h} \sum_{t=h+1}^{T} \mathbb{1}\{y_t \in \hat{\mathscr{C}}^{\alpha}_{t|t-h}\} - (1-\alpha) \right| \leq 2\frac{c \cdot g(T-h) + h}{T-h} + \chi_{T,h}, \tag{15}$$

so the empirical coverage converges to $1-\alpha$ whenever $\chi_{T,h} \to 0$.

Proof. See Appendix A8. □

Unlike the preceding single-tail results, Corollary 3 provides a lower bound on empirical coverage of intervals and ensures a limit inferior of at least $1-\alpha$, while convergence to exactly $1-\alpha$ follows when the endpoint-crossing frequency vanishes. Setting $\hat{q}_{t+h|t} = 0$, the result also covers the integral-only iteration in Proposition 3. Analogous two-sided guarantees for the constant-step quantile-tracking update shown in Corollary 1 are given in Corollary 4 in Appendix A9.

These results bound empirical coverage at each forecast horizon, both in finite samples and in the long run, but do not ensure conditional coverage at individual forecast origins or simultaneous coverage across horizons. No method can provide such conditional guarantees in finite samples without distributional assumptions: even for i.i.d. data, a procedure with distribution-free conditional coverage must produce intervals of infinite expected length whenever the covariate distribution is nonatomic (Lei & Wasserman 2014, Barber et al. 2021), and our setting includes

i.i.d. data as a special case. Nor do they guarantee nominal coverage in every local window. Appendix A10 shows that the coverage shortfall over a window is bounded by twice the sum of the cumulative calibration bounds at its endpoints, divided by the window length. This bound vanishes when the window length grows faster than that sum, but may remain uninformative for fixed-length rolling windows. Thus, long-run calibration alone does not guarantee protection against local undercoverage.

5 Experiments

We evaluate MPID and AcMCP using two simulated settings and two real-world data sets. For benchmarks, we use the multi-step extensions of split conformal prediction (MSCP), weighted conformal prediction (MWCP), and adaptive conformal prediction (MACP) described in Appendix A1, together with the multi-step sequential predictive conformal inference (SPCI) proposed by Xu & Xie (2023). The following parameter choices are used:

(a) we target coverage $1-\alpha = 0.9$, with equal tail miscoverage levels $\alpha^- = \alpha^+ = 0.05$;

(b) for MWCP, we use $b = 0.99$ as per Barber et al. (2023);

(c) following Angelopoulos et al. (2023), MACP uses a step size $\gamma = 0.005$; MPID uses a Theta model as the scorecaster; and both MPID and AcMCP use a learning rate of $\eta = 0.1\hat{B}_t$ for quantile tracking, where $\hat{B}_t = \max\{s_{t-\Delta+1|t-\Delta-h+1}, \ldots, s_{t|t-h}\}$ is the highest score over a trailing window of length Δ, which is set equal to the calibration set size. Because this learning rate varies over time, Corollary 1, which assumes a constant step, does not apply; however, $\hat{B}_t$ depends only on past scores, so the P state remains a function of the past and the guarantees of Corollaries 2 and 3 continue to hold;

(d) we adopt a nonlinear saturation function $r_t(x) = K_{\mathrm{I}} \tan\left(x \log(t)/(tC_{\mathrm{sat}})\right)$, where $\tan(x) = \mathrm{sign}(x) \cdot \infty$ for $x \notin [-\pi/2, \pi/2]$, and constants $C_{\mathrm{sat}}, K_{\mathrm{I}} > 0$ are chosen heuristically as in Angelopoulos et al. (2023);

(e) we use a clipped version of MACP, replacing an infinite upper (lower) bound by the point

forecast plus (minus) the largest upper-tail (lower-tail) score observed by the forecast origin;

(f) for SPCI, we follow the implementation of Xu & Xie (2023), using a conditional quantile forest with 10 trees and maximum depth 2. For each horizon, we refit it over the rolling calibration window using the latest 10 available signed errors as predictors and construct intervals at fixed quantile levels 0.05 and 0.95. All methods share point forecasts and out-of-sample calibration errors.

5.1 Simulated linear autoregressive process

We first consider a simulated stationary time series generated from a simple AR(2) process

$$y_t = 0.8y_{t-1} - 0.5y_{t-2} + \varepsilon_t,$$

where ε_t is white noise with error variance $\sigma^2 = 1$. After an appropriate burn-in period, we generate $N = 5000$ data points. Under the sequential split and online learning settings, we create training sets $\mathscr{D}_{\text{tr}}$ and calibration sets $\mathscr{D}_{\text{cal}}$, each with a length of 500. We use AR(2) models to generate 1- to 3-step-ahead point forecasts (i.e. $H = 3$), using the `Arima()` function from the `forecast` R package (Hyndman & Khandakar 2008, Hyndman et al. 2026). The goal is to generate prediction intervals using various proposed conformal prediction methods and evaluate whether they can achieve the nominal long-run coverage for each separate forecast horizon.

Figure 2 presents the rolling coverage and mean interval width of each method for each forecast horizon, computed over rolling windows of size 500; rolling median widths behave similarly and are omitted. The analytic intervals from the fitted AR model are also shown. MPID and AcMCP stay close to the 90% target across windows, whereas the other methods, including the AR model, swing much further from it. AcMCP adapts its interval widths to past coverage, widening them when competing methods undercover. It attains coverage similar to MPID with narrower intervals, an advantage that grows with h without deteriorating at smaller h. Figure 3 presents the corresponding boxplots of rolling coverage and interval width.

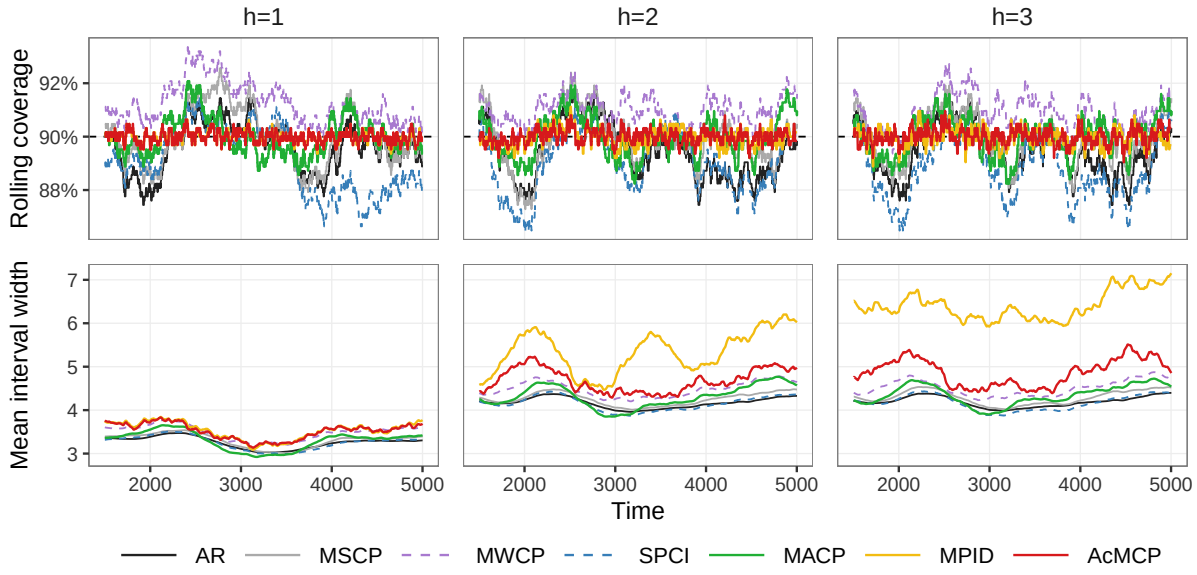


Figure 2: *AR(2) simulation results showing rolling coverage (top) and mean interval width (bottom) for each forecast horizon. Both metrics are computed over rolling windows of size* 500*. The black dashed horizontal line indicates the target coverage level* $1-\alpha = 0.9$*.*

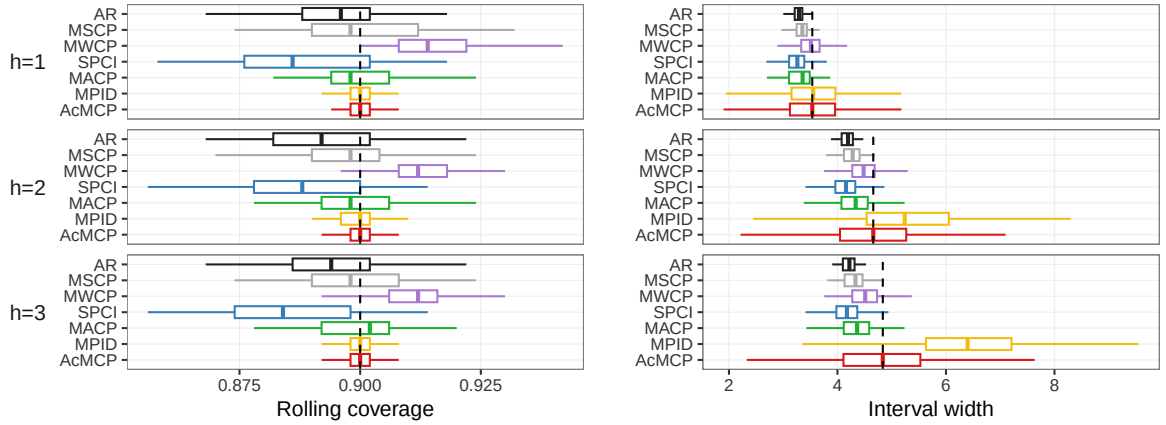


Figure 3: *AR(2) simulation results showing boxplots of the rolling coverage and interval width for each method across different forecast horizons. Black dashed lines indicate the target coverage level in the left panels and the median interval width of AcMCP in the right panels.*

To examine the effect of the scorecasting term $\tilde{s}_{t+h|t}$, we compare AcMCP with the version without this term, denoted by MPI. Figure 4 displays their prediction intervals over a selected period of 500 observations. The two sets of bounds mostly overlap; the main differences occur around time 1320 and during 1470–1500, where AcMCP fans out locally relative to MPI, most

visibly at $h = 3$. The figure also shows MACP, which likewise updates its intervals using past miscoverage. AcMCP and MACP both capture some patterns in the observations, and neither consistently yields narrower intervals.

5.2 Simulated nonlinear autoregressive process

Next, consider the case of a nonlinear data-generating process, defined as

$$y_t = \sin(y_{t-1}) + 0.5\log(y_{t-2}+1) + 0.1y_{t-1}x_{1,t} + 0.3x_{2,t} + \varepsilon_t,$$

where $x_{1,t}$ and $x_{2,t}$ are uniformly distributed on $[0,1]$, and ε_t is white noise with variance $\sigma^2 = 0.01$. Thus, y_t nonlinearly depends on its lagged values y_{t-1}, y_{t-2}, and exogenous variables $x_{1,t}$ and $x_{2,t}$.

After an appropriate burn-in period, we generate $N = 2000$ data points. Under the sequential split and online learning settings, both the training set $\mathscr{D}_{\text{tr}}$ and the calibration set $\mathscr{D}_{\text{cal}}$ have size 500. Given the nonlinear DGP, we use feed-forward neural networks with a single hidden layer and lagged inputs to generate 1- to 3-step-ahead point forecasts ($H = 3$), using the `nnetar()`

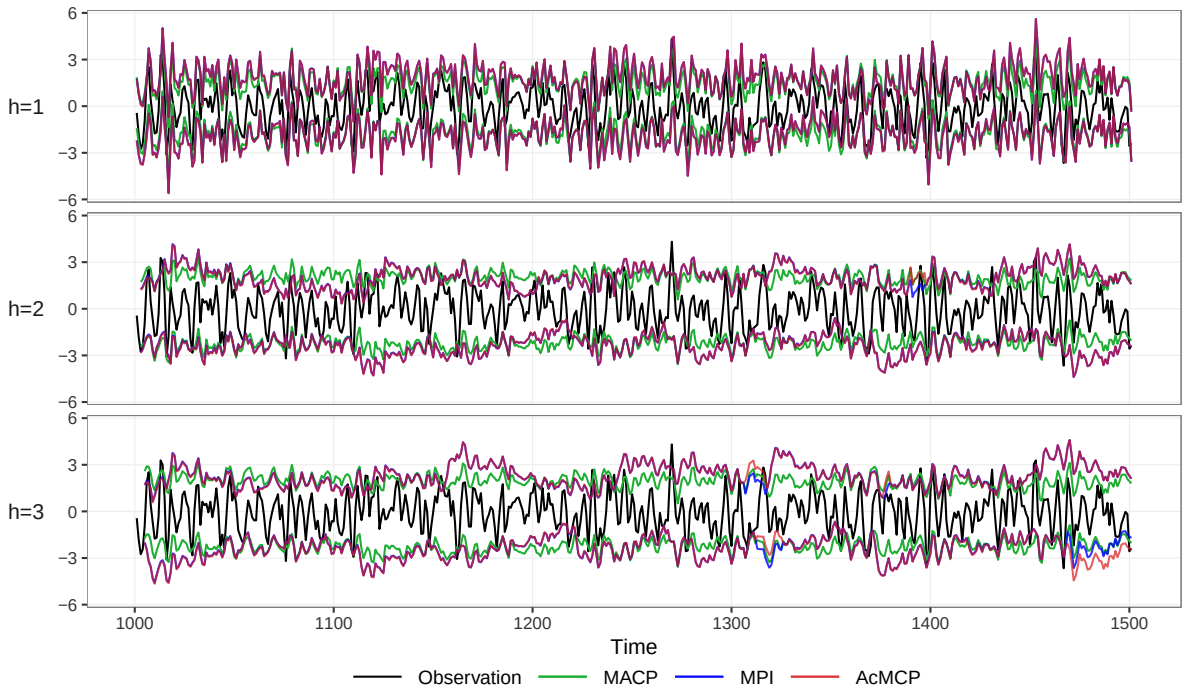


Figure 4: *AR(2) simulation results showing the prediction interval bounds for the MACP, MPI, and AcMCP methods over a truncated period of length 500.*

function from the `forecast` R package (Hyndman et al. 2026). Note that analytic prediction intervals are unavailable for neural networks, so we do not include them when presenting the results.

Figure 5 shows the rolling coverage and mean interval width of each method over rolling

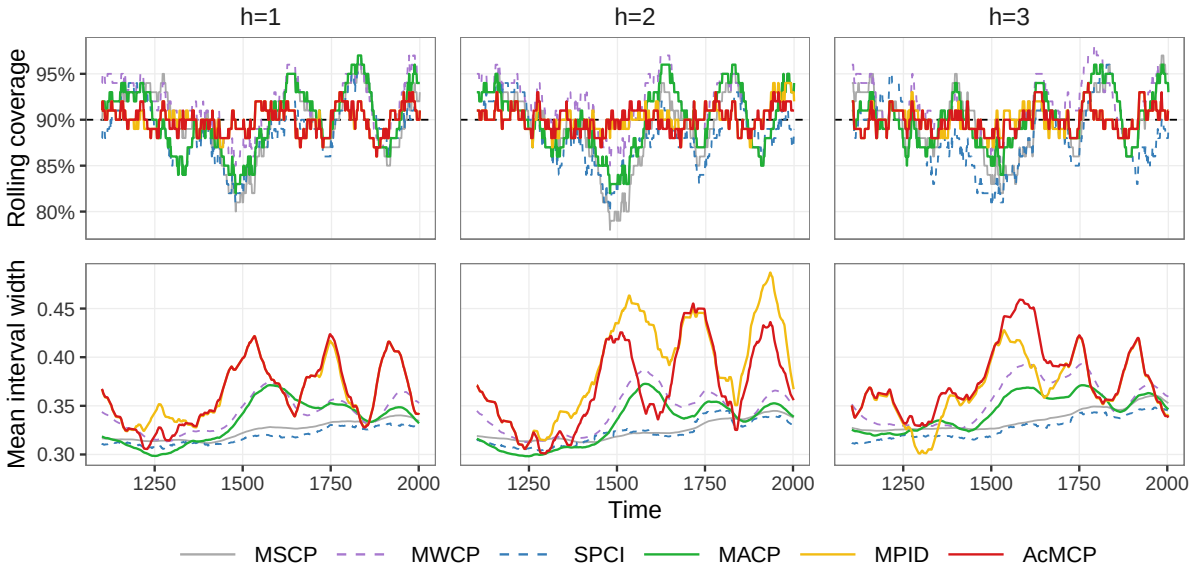


Figure 5: *Nonlinear simulation results showing rolling coverage (top) and mean interval width (bottom) for each forecast horizon. Both metrics are computed over rolling windows of size* 100*. The black dashed horizontal line indicates the target coverage level* $1 - \alpha = 0.9$.

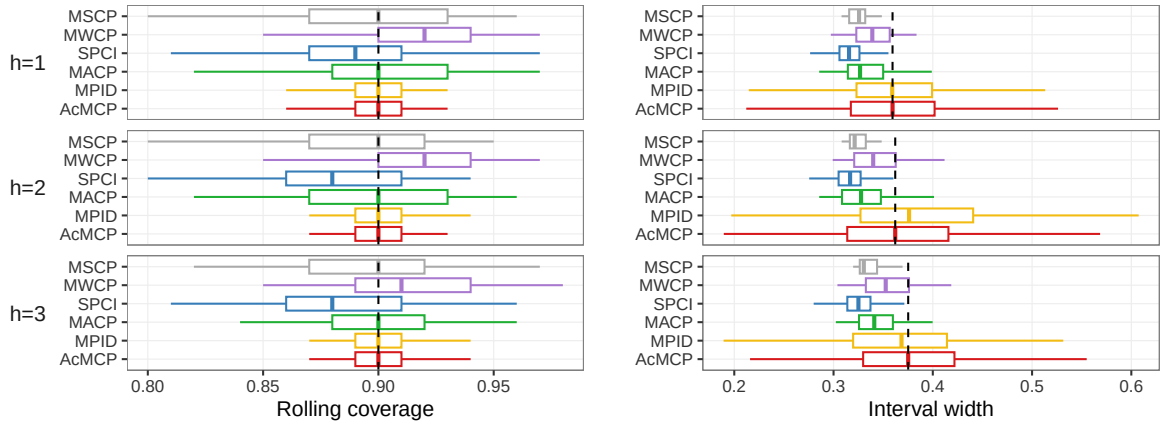


Figure 6: *Nonlinear simulation results showing boxplots of the rolling coverage and interval width for each method across different forecast horizons. Black dashed lines indicate the target coverage level in the left panels and the median interval width of AcMCP in the right panels.*

windows of size 100. MPID and AcMCP fluctuate only slightly around the 90% target, whereas MSCP, MWCP, SPCI and MACP show pronounced fluctuations. To maintain coverage, MPID and AcMCP vary their interval widths more than the competing methods. AcMCP generally produces narrower intervals than MPID at $h = 2$ but wider intervals at $h = 3$. Under AcMCP, the interval width at $h = 3$ exceeds that at $h = 2$ in 53.41% of the test cases, compared with 38.08% under MPID, which is more consistent with forecast uncertainty increasing with the horizon. The boxplots in Figure 6 confirm that MPID and AcMCP have less variable rolling coverage than MSCP, MWCP, SPCI and MACP, but larger and more variable interval widths.

5.3 Electricity demand data

We apply the conformal prediction methods using data comprising daily electricity demand (GW), daily maximum temperature (degrees Celsius), and holiday information for Victoria, Australia, from 2012 to 2014. Temperatures are taken from the Melbourne Regional Office weather station. The left panel of Figure 7 displays the daily electricity demand and temperatures over the period. The right panel shows a nonlinear relationship between demand and temperature, with demand increasing for low temperatures (due to heating) and increasing for high temperatures (due to cooling); the two clouds of points correspond to working days and non-working days.

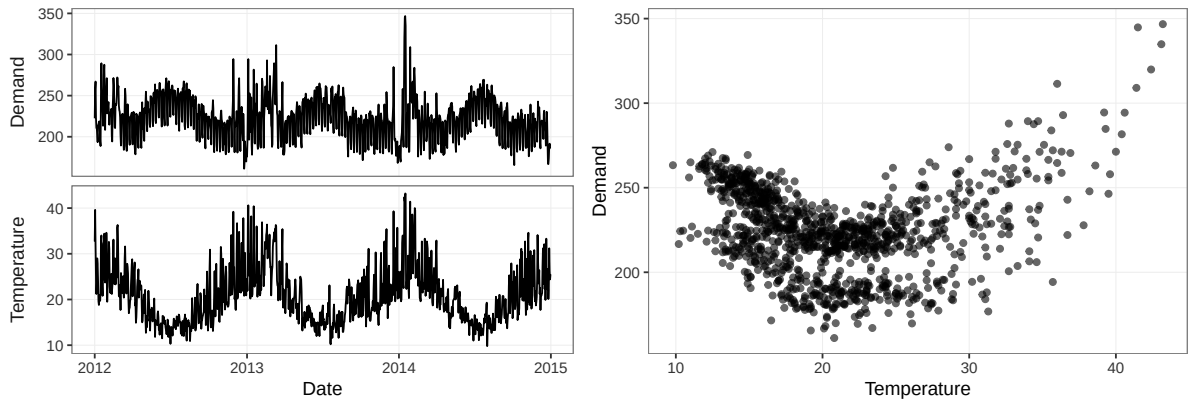


Figure 7: *Daily electricity demand and corresponding daily maximum temperatures in 2012–2014, Victoria, Australia.*

The response variable is `Demand`, with two covariates: `Temperature` and `Workday` (an indicator

variable for working days). Following Hyndman & Athanasopoulos (2021), we fit a dynamic regression model with a piecewise linear function of temperature (with a knot at 18 degrees) to generate 1- to 7-step-ahead point forecasts ($H = 7$), modeling the errors with an ARIMA process to capture autocorrelation. We use two years of data for training and 100 observations for calibration.

Figure 8 and Figure 9 compare rolling coverage and interval width across methods over rolling windows of size 100; DR denotes the analytic intervals from the dynamic regression model. DR consistently overcovers, giving much wider intervals than the other methods for $h \leq 4$. MSCP, MWCP and MACP fluctuate substantially in rolling coverage and undercover in many windows after September 2014 at all horizons. SPCI produces narrower intervals than MPID and AcMCP, but at the cost of substantial undercoverage across horizons. MPI, MPID and AcMCP give wider intervals that largely avoid the post-September 2014 undercoverage; among these, MPID has slightly poorer coverage than MPI and AcMCP at $h = 3$ despite wider intervals. AcMCP has coverage similar to MPI but narrower intervals, particularly at longer horizons, and smaller mean interval widths than MPID at every horizon, with reductions of 18.94% at $h = 3$ and 22.29% at $h = 7$.

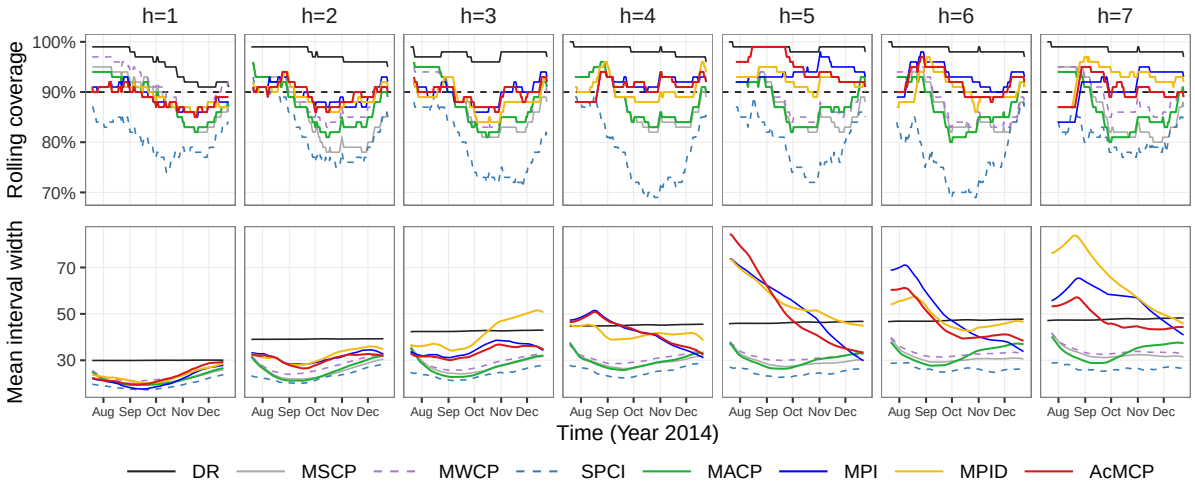


Figure 8: *Electricity demand data results showing rolling coverage (top) and mean interval width (bottom) for each forecast horizon. Both metrics are computed over rolling windows of size* 100*. The black dashed horizontal line indicates the target coverage level* $1 - \alpha = 0.9$.

AcMCP achieves overall coverage close to the nominal level at most horizons, although it overcovers at $h = 5$, and has smaller mean and median interval widths than MPID at every horizon. These properties support its use in short-term generation planning: combined with supply forecasts, the intervals can flag days that warrant closer review of capacity margins and support assessments of candidate generation plans under different demand scenarios. However, satisfactory overall and rolling coverage does not imply conditional coverage under extreme temperatures, high demand, or other critical operating conditions. The theoretical guarantees concern empirical coverage at each horizon; they ensure neither coverage conditional on the information available for a given

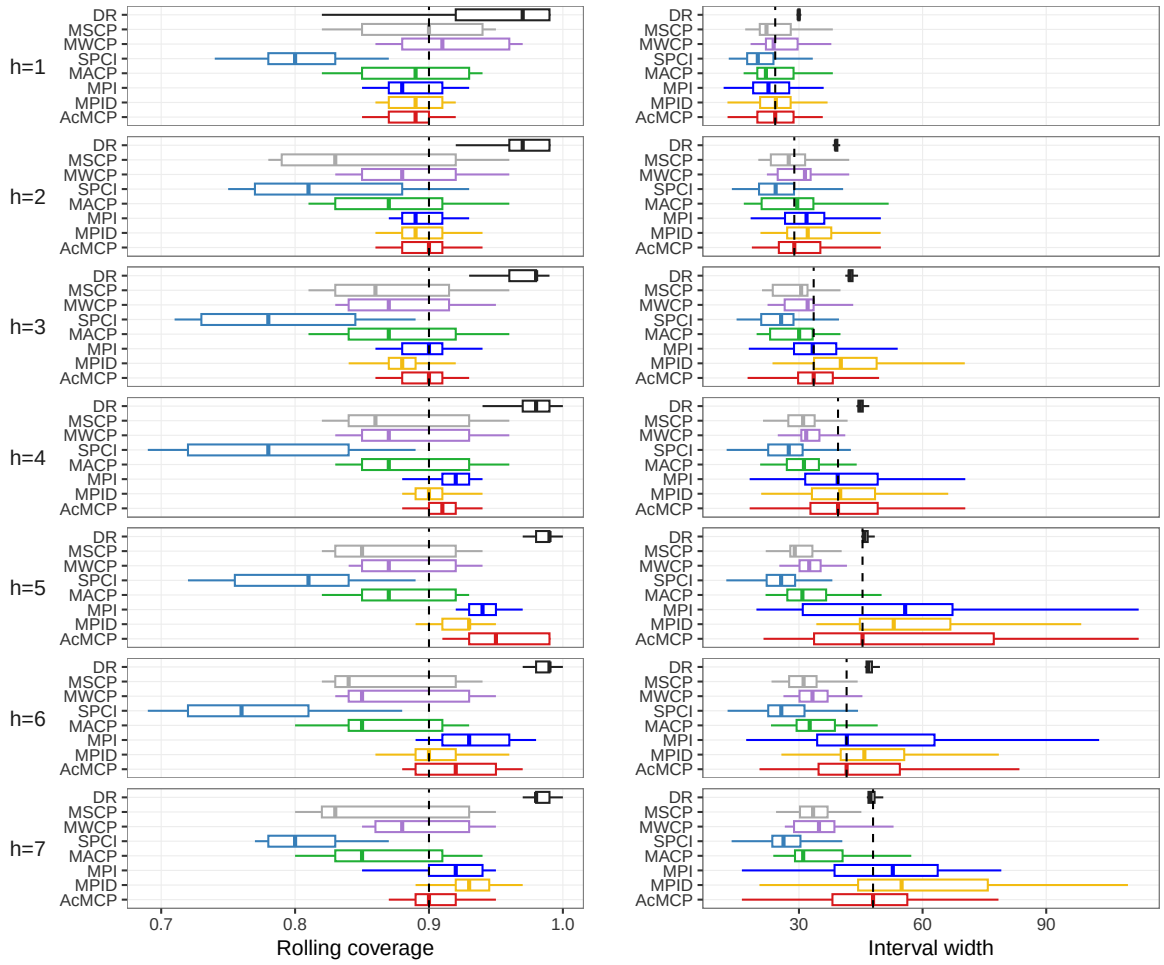


Figure 9: *Electricity demand data results showing boxplots of the rolling coverage and interval width for each method across different forecast horizons. Black dashed lines indicate the target coverage level in the left panels and the median interval width of AcMCP in the right panels.*

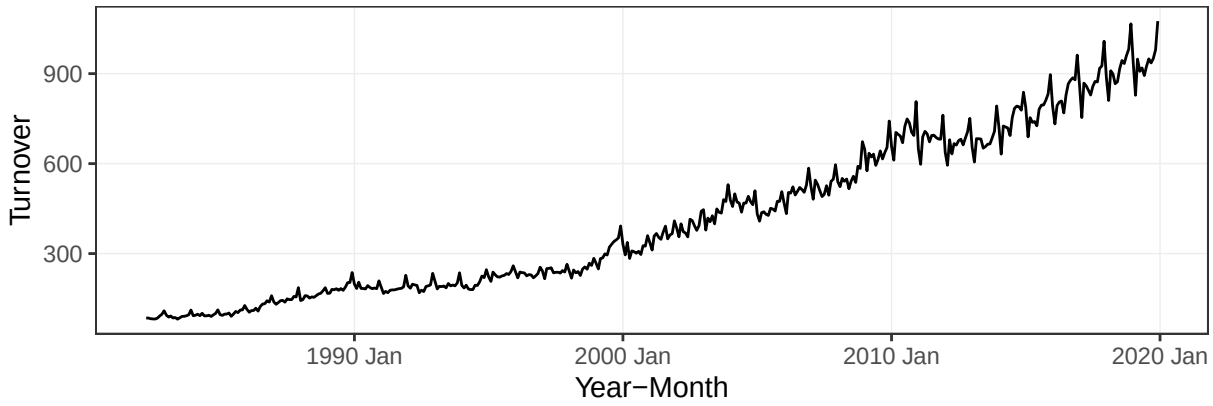


Figure 10: *Monthly expenditure on cafes, restaurants and takeaway food services in Victoria, Australia, from April 1982 to December 2019.*

decision nor simultaneous coverage across horizons. Costly upper-tail exceedances can therefore occur despite satisfactory average coverage. Before using the intervals to set reserves, practitioners should assess the frequency and magnitude of upper-bound exceedances during high-demand periods, and weigh this exposure against available generation capacity and the cost of covering any shortfall.

5.4 Eating out expenditure data

In our final example, we apply the conformal prediction methods to forecast the eating out expenditure (in millions of dollars) in Victoria, Australia. The data include monthly expenditure on cafes, restaurants and takeaway food services from April 1982 to December 2019, as shown in Figure 10. The data show an overall upward trend, clear annual seasonality, and level-dependent variability.

We consider three models: ARIMA with logarithmic transformation, ETS, and STL-ETS (Hyndman & Athanasopoulos 2021), and then output their simple average as final point forecasts. STL-ETS applies ETS to the seasonally adjusted series obtained via STL decomposition. All models are automatically trained using the `forecast` R package (Hyndman et al. 2026). We aim to forecast 12 months ahead ($H = 12$) using 20 years of data for training and 5 years for

calibration. The test period contains 153 months for one-step evaluation, with fewer observed targets at longer horizons. We therefore summarize coverage and interval width over the available evaluation periods rather than showing rolling windows.

Because the horizons in this application are relatively long, Figure 11 reports the coverage gap and mean interval width over the whole test set for each method and horizon. SPCI produces narrower intervals but undercovers at every horizon, especially longer ones. MSCP, MWCP and MACP are conservative at short horizons, particularly MWCP and MACP, but all three undercover for $h \geq 8$; overall, the coverage of MSCP, MWCP, SPCI and MACP declines as the horizon increases. MPI and AcMCP approximately achieve the target coverage for $h \leq 5$ with mean widths comparable to those of MSCP, MWCP and MACP, and keep coverage within a tighter range across horizons, albeit at the cost of interval efficiency. AcMCP has smaller mean interval widths than MPID at all twelve horizons.

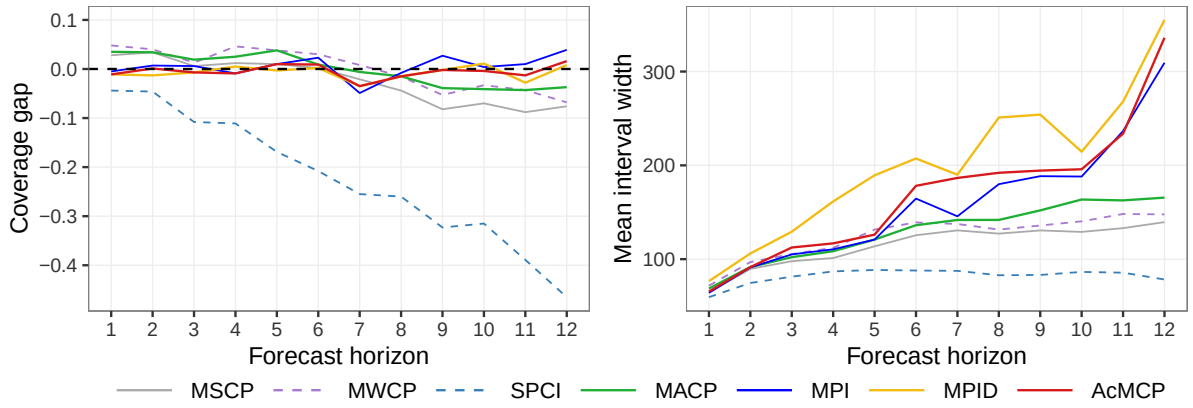


Figure 11: *Eating out expenditure data results showing empirical coverage minus the* 90% *target level (left) and mean interval width (right), computed over the entire test set for each forecast horizon. The black dashed line indicates zero coverage gap.*

5.5 Post-hoc comparison

For MACP, the step size γ controls both the calibration tightness and the responsiveness of the interval widths: smaller values lead to smoother interval evolution but slower correction of local coverage errors, while larger values improve responsiveness at the cost of increased variability.

An analogous trade-off arises in AcMCP, where the learning rate η governs the balance between adaptivity and stability. This motivates a comparison with MACP tuned for comparable local coverage variability.

To examine this, we conduct a post-hoc comparison in which the learning-rate rule and tuning constants for AcMCP are held fixed, while γ for MACP is selected by grid search over $[0.005, 0.5]$ in steps of 0.015. Specifically, we choose the smallest γ for which the sample variance of one-step-ahead local coverage over rolling windows of the test set does not exceed that of AcMCP, and use it at all horizons. This is a post-hoc comparison based on the test set, which favors MACP, rather than a prospective tuning procedure or a guarantee of optimality. The selected step sizes, $\gamma = 0.065$, 0.05, 0.035 and 0.035 for the four data sets, are much larger than the default $\gamma = 0.005$. The working miscoverage level of MACP therefore often falls low enough to produce infinite bounds, which are clipped as described in Section 5; depending on the data set and horizon, approximately 12% to 89% of the issued tuned MACP intervals have at least one bound clipped.

Table 1 summarizes, for each forecast horizon, mean coverage on observed targets, the minimum and maximum local coverage across rolling windows, and the mean and median widths of issued intervals for MACP and AcMCP. In the AR(2) simulation, both methods attain mean coverage close to the nominal 90% with similar ranges of local coverage, and AcMCP produces narrower intervals at every horizon. In the nonlinear simulation, both methods achieve near-nominal coverage. Neither method consistently yields smaller mean interval widths across horizons, whereas MACP consistently has slightly smaller median widths.

For electricity demand, tuned MACP achieves near-nominal overall coverage but exhibits local undercoverage, with minimum local coverage as low as 84%. AcMCP has higher minimum local coverage at most horizons, although it overcovers at $h = 5$. AcMCP also has smaller mean interval widths at six horizons, with reductions of up to approximately 19%. MACP has smaller median widths at most horizons, but occasional very wide intervals inflate its mean widths even after clipping. For eating out expenditure, the clearest differences arise at longer horizons. AcMCP has

overall coverage closer to the target for $h \geq 9$ and higher minimum local coverage for $h \geq 8$, with the latter improving by approximately 10–18.3 percentage points. However, AcMCP intervals are considerably wider for $h \geq 6$. This additional width is not always accompanied by better overall coverage: at $h = 6$ and $h = 8$, the two methods have similar overall coverage, while at $h = 7$ MACP attains higher coverage with narrower intervals. Overall, tuning the step size substantially narrows the gap between MACP and AcMCP. AcMCP, which is not tuned on the test set, retains more uniform coverage across horizons, but its intervals are not uniformly shorter: at longer horizons in the real data sets, they can be wider than those of a tuned MACP with similar coverage.

Table 1: *Performance of MACP and AcMCP across the four datasets, summarized by mean test-set coverage, minimum and maximum local coverage over rolling windows, and mean and median interval widths for each forecast horizon.*

	Mean coverage (%)		Min. local coverage (%)		Max. local coverage (%)		Mean interval width		Median interval width	
	MACP	AcMCP	MACP	AcMCP	MACP	AcMCP	MACP	AcMCP	MACP	AcMCP
					AR(2) simulation data					
h=1	90.00	90.00	89.20	89.40	90.80	90.80	3.83	3.55	3.63	3.54
h=2	89.99	89.94	89.20	89.00	91.00	91.00	5.17	4.68	4.91	4.66
h=3	89.99	90.02	88.80	88.80	91.00	91.20	5.32	4.86	5.23	4.84
					Nonlinear simulation data					
h=1	90.00	90.00	87.00	86.00	93.00	93.00	0.35	0.36	0.33	0.36
h=2	89.98	90.18	87.00	87.00	94.00	93.00	0.41	0.37	0.34	0.36
h=3	89.96	90.06	87.00	87.00	94.00	94.00	0.38	0.38	0.36	0.38
					Electricity demand data					
h=1	90.31	89.15	86.00	85.00	92.00	92.00	26.18	24.44	22.46	24.24
h=2	90.62	90.23	85.00	86.00	94.00	94.00	37.87	30.72	30.92	28.87
h=3	90.55	90.94	85.00	86.00	93.00	93.00	42.03	34.35	30.93	33.57
h=4	90.87	90.87	86.00	88.00	96.00	94.00	43.24	40.74	33.79	39.48
h=5	91.20	94.80	87.00	91.00	96.00	99.00	64.01	55.62	55.19	45.43
h=6	90.73	90.73	84.00	88.00	94.00	97.00	45.89	46.65	39.39	41.56
h=7	90.24	89.84	85.00	87.00	93.00	95.00	50.81	47.45	35.81	47.98
					Eating out expenditure data					
h=1	92.16	88.89	90.00	83.33	96.67	93.33	67.49	65.51	64.89	63.08
h=2	92.05	90.07	85.00	83.33	96.67	96.67	90.12	91.50	81.98	89.10
h=3	91.28	89.26	86.67	83.33	96.67	95.00	105.01	112.42	96.03	107.83
h=4	91.84	89.12	86.67	81.67	98.33	98.33	132.90	116.80	120.67	111.16
h=5	92.41	91.03	85.00	81.67	98.33	96.67	133.90	126.12	115.52	120.97
h=6	91.61	90.91	85.00	85.00	98.33	96.67	135.21	178.17	125.62	178.60
h=7	88.65	86.52	78.33	73.33	98.33	98.33	140.76	186.58	140.97	198.27
h=8	87.77	88.49	71.67	83.33	100.00	98.33	148.63	192.07	148.90	192.59
h=9	86.13	89.78	68.33	86.67	100.00	100.00	156.46	194.40	161.81	168.96
h=10	88.15	89.63	75.00	85.00	100.00	93.33	166.56	195.83	170.70	193.68
h=11	85.71	88.72	70.00	85.00	100.00	96.67	163.92	233.80	158.95	220.92
h=12	85.50	91.60	68.33	83.33	100.00	98.33	168.05	336.02	181.64	297.48

6 Conclusion and discussion

We propose MPID and AcMCP to construct prediction sets for multi-step time series forecasting. Both methods extend conformal PID control (Angelopoulos et al. 2023) to accommodate delayed

feedback inherent in this setting, while AcMCP additionally incorporates dependence among forecast errors across horizons into its scorecaster. Decompositions of optimal forecast errors, together with explicit bounds on Taylor remainders, provide a structural rationale for this design and clarify the conditions supporting the underlying linear approximations. For both methods, we establish finite-sample bounds and horizon-specific long-run guarantees for empirical coverage under the stated calibration conditions, without requiring exchangeability, stationarity, or assumptions on how the data distribution evolves over time. These guarantees do not depend on the accuracy of the linear approximations. The coverage bounds distinguish the contribution of calibration feedback from that of forecast delay, whose effect vanishes asymptotically for each fixed horizon. Across the experiments considered, MPID and AcMCP adapt their interval widths to changing time series dynamics, generally maintaining more stable rolling coverage around the target than the benchmarks. AcMCP often produces narrower intervals than MPID with similar coverage, indicating the empirical value of incorporating forecast error dependence into conformal calibration. This advantage is not uniform, however: when the step size of MACP is tuned post hoc on the test set, MACP produces intervals comparable to those of AcMCP and, at longer horizons in the real data sets, sometimes narrower ones with similar coverage.

Our analysis of forecast errors is restricted to an ex-post forecasting setting, in which future values of exogenous predictors are assumed to be known. In this setting, interval width reflects both *aleatoric uncertainty*, arising from intrinsic randomness in the DGP, and *epistemic uncertainty*, stemming from a lack of knowledge about the underlying DGP (Sale et al. 2025). In many practical forecasting scenarios, however, predictors must themselves be forecast. Uncertainty in future predictor values and in the models used to forecast them can then propagate into response forecasts, motivating explicit treatment of predictor uncertainty in conformal inference.

Although the forecast error analysis assumes known future predictors, the coverage guarantees remain applicable when these predictors are forecast, provided that the assumptions of Corollary 2 and Corollary 3 hold. Appendix A11 compares AcMCP using known and forecast future predictors

in a simulation. Both settings achieve near-nominal empirical coverage and small rolling coverage errors, but forecasting the predictors produces wider intervals, with larger increases at higher levels of predictor uncertainty and longer forecast horizons.

These considerations suggest several directions for future research. One is to improve interval efficiency while preserving coverage guarantees. Our methods prioritize efficient online implementation through simple updates. A complementary approach is Bellman conformal inference (Yang, Candès & Lei 2024), which uses a look-ahead optimization to balance interval length against a miscoverage penalty, but issues only a one-step-ahead interval. Extending this optimization principle to multi-step interval construction while retaining coverage guarantees is a promising direction. Automated learning-rate selection could further reduce manual tuning and improve adaptation to changing forecasting conditions. Other avenues include establishing conditional coverage guarantees under explicit restrictions on distributional change and developing a theoretical understanding of how uncertainty in future predictors and their forecasting models affects interval efficiency and local calibration.

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Disclosure statement

The authors report there are no competing interests to declare.

Supplementary materials

The Appendix provides proofs of all propositions and corollaries, additional multi-step conformal methods, analyses of approximation errors and local coverage, supplementary two-sided coverage results, and a simulation examining the effect of uncertainty in future predictors.

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Appendix

A1 Multi-step extensions of existing conformal methods

A1.1 Online multi-step split conformal prediction

Split conformal prediction (SCP, Papadopoulos et al. 2002, Vovk et al. 2005) is a holdout method for building prediction sets using a model pre-trained on a training set. A key advantage of SCP is its ability to guarantee coverage by assuming data exchangeability. In a regression setting, SCP randomly separates the available n data points, $Z_i = (X_i, Y_i) \in \mathbb{R}^d \times \mathbb{R}$, $i = 1, \ldots, n$, into a proper training set of size n_t and a calibration set of size n_c. Given a regression model $\hat{\mu} : \mathbb{R}^d \to \mathbb{R}$ fitted on the training set and a score function $\mathscr{S}$, nonconformity scores $s_i = \mathscr{S}(X_i, Y_i)$ are computed on the calibration set to measure the nonconformity between the calibration response values and the predicted values obtained from the fitted model $\hat{\mu}$. Then SCP computes the prediction set for the test response Y_{n+1} using

$$\hat{\mathscr{C}}^{\alpha}_{n+1}(X_{n+1}) = \left\{ y \in \mathbb{R} : \mathscr{S}(X_{n+1}, y) \leq Q_{1-\alpha}\left(\sum_{i \in \mathscr{D}_{\text{cal}}} \frac{1}{n_c+1} \cdot \delta_{s_i} + \frac{1}{n_c+1} \cdot \delta_{+\infty} \right) \right\},$$

where $Q_\tau(\cdot)$ denotes the τ-quantile of its argument, and δ_a denotes the point mass at a. Time series data are inherently nonexchangeable due to their temporal dependence and autocorrelation. Therefore, directly applying SCP to time series data would violate the method's exchangeability assumption and compromise its coverage guarantee.

Here we introduce online **multi-step split conformal prediction** (MSCP) as a generalization of SCP (Papadopoulos et al. 2002, Vovk et al. 2005) to recursively update all h-step-ahead prediction intervals over time. MSCP applies conformal inference in an online fashion, updating prediction intervals as new data points are received. Specifically, for each $h \in [H]$, the one-sided prediction set is

$$\hat{\mathscr{C}}^{\alpha,\text{tail}}_{t+h|t} = \left\{ y \in \mathbb{R} : s^{y}_{t+h|t} \leq Q_{1-\alpha}\left(\sum_{i=t-t_c+1}^{t} \frac{1}{t_c+1} \cdot \delta_{s_{i|i-h}} + \frac{1}{t_c+1} \cdot \delta_{+\infty} \right) \right\}, \tag{16}$$

where $s^y_{t+h|t}$ is the relevant signed nonconformity score defined in Section 2. Applying Equation (16) to both tails and combining the resulting thresholds as in Equation (2) yields the MSCP interval.

A1.2 Online multi-step weighted conformal prediction

Barber et al. (2023) propose a nonexchangeable conformal prediction procedure (NexCP) that generalizes SCP to certain nonexchangeable settings by assigning higher weights to observations believed to share the same distribution as the test data. NexCP assumes these weights are fixed and data-independent. Under exchangeability, NexCP retains the same coverage guarantees as SCP; when exchangeability is violated, the coverage gap is governed by the total variation distance between swapped nonconformity score vectors, which can be substantial in time series settings.

The online **multi-step weighted conformal prediction** (MWCP) method we propose here adapts the NexCP method to the online setting for time series forecasting. MWCP uses weighted quantile estimates for constructing prediction intervals, contrasting with the MSCP definitions where all nonconformity scores for calibration are implicitly assigned equal weight.

For the subsequent empirical study, we choose fixed weights $w_i = b^{t+1-i}$, $b \in (0,1)$ and $i = t - t_c + 1, \ldots, t$, for nonconformity scores on the corresponding calibration set. In this setting, weights decay exponentially as the nonconformity scores get older, akin to the rationale behind the simple exponential smoothing method in time series forecasting (Hyndman & Athanasopoulos 2021). Then, for each $h \in [H]$, MWCP updates the h-step-ahead one-sided prediction set:

$$\hat{\mathscr{C}}^{\alpha,\text{tail}}_{t+h|t} = \left\{ y \in \mathbb{R} : s^y_{t+h|t} \leq Q_{1-\alpha}\left(\sum_{i=t-t_c+1}^{t} \tilde{w}_i \cdot \delta_{s_{i|i-h}} + \tilde{w}_{t+1} \cdot \delta_{+\infty} \right) \right\},$$

where $\tilde{w}_i$ and $\tilde{w}_{t+1}$ are normalized weights given by

$$\tilde{w}_i = \frac{w_i}{\sum_{i=t-t_c+1}^{t} w_i + 1}, \text{ for } i \in \{t - t_c + 1, \ldots, t\} \quad \text{and} \quad \tilde{w}_{t+1} = \frac{1}{\sum_{i=t-t_c+1}^{t} w_i + 1}.$$

As with MSCP, the two one-sided MWCP sets are combined according to Equation (2).

A1.3 Multi-step adaptive conformal prediction

Next we extend the adaptive conformal inference (ACI) method proposed by Gibbs & Candès (2021) to address multi-step time series forecasting, introducing the **multi-step adaptive conformal prediction** (MACP) method. For a generic tail with fixed target miscoverage level α, the method adapts the nominal level $\alpha_{t+h|t}$ used for quantile estimation to past tail miscoverage indicators. Specifically, for each $h \in [H]$, we update $\alpha_{t+h|t}$ according to

$$\alpha_{t+h|t} := \alpha_{t+h-1|t-1} + \gamma(\alpha - \mathrm{err}_{t|t-h}). \tag{17}$$

The corresponding one-sided set is computed from Equation (16) with α replaced by $\alpha_{t+h|t}$. For levels at or below zero, the one-sided set is $\mathbb{R}$; for levels at or above one, it is empty. Here, $\gamma > 0$ is a fixed step size, and $\alpha_{2h|h}$ is the initial estimate, typically set to α. The upper- and lower-tail recursions are run separately and then combined through Equation (2).

Equation (17) uses the tail miscoverage indicator observed at time t to update the forecast for time $t+h$. The nominal tail miscoverage level increases when the observation satisfies the relevant upper or lower bound ($s_{t|t-h} \leq q_{t|t-h}$) and decreases when it violates that bound ($s_{t|t-h} > q_{t|t-h}$). For a fixed calibration sample, decreasing this level widens the one-sided prediction set. Equation (17) reduces to the ACI update when $h = 1$.

We do not use the update $\alpha_{t+1|t-h+1} := \alpha_{t|t-h} + \gamma(\alpha - \mathrm{err}_{t|t-h})$, because it requires y_t, which is unavailable at the forecast origin $t-h+1$ when $h > 1$.

Selecting the parameter γ is pivotal yet challenging. Gibbs & Candès (2021) relate the choice of γ to the rate of distributional change. Several strategies have been proposed to avoid the necessity of selecting γ. For example, Zaffran et al. (2022) use an adaptive aggregation of multiple ACIs with a set of candidate values for γ, determining weights based on their historical performance. Bastani et al. (2022) propose a multivalid prediction algorithm in which the prediction set is established by selecting a threshold from a sequence of candidate thresholds. Both methods rely

on update schemes that place substantial weight on older historical data, which may impede rapid adaptation to abrupt changes (Gibbs & Candès 2024). Therefore, Gibbs & Candès (2024) propose an alternative expert selection scheme that adaptively tunes the step-size parameter over time and places greater emphasis on more recent observations by construction, enabling faster responses to sudden environmental shifts.

The theoretical coverage properties of ACI suggest that a larger value for γ generally results in less deviation from the target coverage. As there is no restriction on $\alpha_{t+h|t}$, it can drift below 0 or above 1; a larger γ may therefore lead to frequent empty or unbounded one-sided sets as the procedure responds to whether the relevant bound is violated.

A2 Proof of Proposition 1

Proof. Fix a forecast origin t. For $h = 1$, all lagged responses and required predictors are $\mathscr{F}_t$-measurable. Since $\mathbb{E}(\varepsilon_{t+1}|\mathscr{F}_t) = 0$, we have

$$\hat{y}_{t+1|t} = f_{t+1}(y_{(t+1-d):t}, \boldsymbol{x}_{(t+1-k):(t+1)})$$

and hence $e_{t+1|t} = \varepsilon_{t+1}$, with $\widetilde{R}_{t,1} = 0$.

For $h \geq 2$, write $\boldsymbol{\Delta}_{t,h} = \boldsymbol{z}_{t,h} - \boldsymbol{m}_{t,h} = (e_{t+h-1|t}, \ldots, e_{t+h-p_h|t})'$. Applying a first-order Taylor expansion and retaining the remainder gives

$$\begin{aligned} y_{t+h} &= F_{t,h}(\boldsymbol{z}_{t,h}) + \varepsilon_{t+h} \\ &= F_{t,h}(\boldsymbol{m}_{t,h}) + \sum_{i=1}^{p_h} \partial_i F_{t,h}(\boldsymbol{m}_{t,h}) e_{t+h-i|t} + R_{t,h} + \varepsilon_{t+h}, \end{aligned}$$

where $R_{t,h}$ is the Taylor remainder beyond first order.

The expansion point and derivative coefficients are $\mathscr{F}_t$-measurable. Since $\mathbb{E}(e_{t+h-i|t}|\mathscr{F}_t) = 0$ for $1 \leq i \leq p_h$ and, by the tower property, $\mathbb{E}(\varepsilon_{t+h}|\mathscr{F}_t) = 0$, taking conditional expectations under the stated integrability assumptions gives

$$\hat{y}_{t+h|t} = F_{t,h}(\boldsymbol{m}_{t,h}) + \mathbb{E}(R_{t,h}|\mathscr{F}_t).$$

Subtracting this identity from the expansion yields

$$e_{t+h|t} = \varepsilon_{t+h} + \sum_{i=1}^{p_h} \phi_{t,h,i} e_{t+h-i|t} + \widetilde{R}_{t,h},$$

where $\widetilde{R}_{t,h} = R_{t,h} - \mathbb{E}(R_{t,h}|\mathscr{F}_t)$ and $\mathbb{E}(\widetilde{R}_{t,h}|\mathscr{F}_t) = 0$. Together with the case $h = 1$, interpreting an empty sum as zero, this proves Equation (7). If f_{t+h} is linear in its lagged responses, allowing an intercept, then $R_{t,h} = \widetilde{R}_{t,h} = 0$.

To establish Equation (8), use the integral representation

$$R_{t,h} = \int_0^1 (1-u)\boldsymbol{\Delta}_{t,h}' \nabla^2 F_{t,h}(\boldsymbol{m}_{t,h} + u\boldsymbol{\Delta}_{t,h})\boldsymbol{\Delta}_{t,h}\, du,$$

where $\nabla^2 F_{t,h}$ is the Hessian with respect to the unobserved response coordinates. The Hessian bound implies $|R_{t,h}| \le (M_{t,h}/2)\|\boldsymbol{\Delta}_{t,h}\|_2^2$. Conditional Jensen's inequality and the triangle inequality give

$$\mathbb{E}\left[|\widetilde{R}_{t,h}|\,|\,\mathscr{F}_t\right] \le 2\mathbb{E}\left[|R_{t,h}|\,|\,\mathscr{F}_t\right] \le M_{t,h}\mathbb{E}\left[\|\boldsymbol{\Delta}_{t,h}\|_2^2\,|\,\mathscr{F}_t\right].$$

Since $\|\boldsymbol{\Delta}_{t,h}\|_2^2 = \sum_{i=1}^{p_h} e_{t+h-i|t}^2$, this proves Equation (8). □

A3 Proof of Proposition 2

Proof. Fix a forecast origin t. By Proposition 1,

$$\begin{aligned} e_{t+1|t} &= \varepsilon_{t+1}, \\ e_{t+j|t} &= \varepsilon_{t+j} + \sum_{i=1}^{p_j} \phi_{t,j,i} e_{t+j-i|t} + \widetilde{R}_{t,j}, \qquad 2 \le j \le h. \end{aligned}$$

Recursively substituting the earlier-horizon relations gives

$$e_{t+h|t} = \varepsilon_{t+h} + \sum_{i=1}^{h-1} \theta_{t,h,i}\varepsilon_{t+h-i} + U_{t,h}.$$

The coefficients $\theta_{t,h,i}$ are finite sums of products of the ϕ coefficients and are therefore $\mathscr{F}_t$-

measurable. Collecting the remainder terms yields

$$U_{t,1} = 0, \qquad U_{t,j} = \widetilde{R}_{t,j} + \sum_{i=1}^{p_j} \phi_{t,j,i} U_{t,j-i}, \qquad 2 \le j \le h. \tag{18}$$

This proves Equation (9) without discarding any Taylor remainder.

Under the additional assumptions of Proposition 2, Equation (8) gives $\mathbb{E}(|\widetilde{R}_{t,j}|\,|\,\mathscr{F}_t) \le MV$ for $2 \le j \le h$. The triangle inequality and $\mathscr{F}_t$-measurability of the coefficients imply

$$\mathbb{E}(|U_{t,j}|\,|\,\mathscr{F}_t) \le MV + \sum_{i=1}^{p_j} |\phi_{t,j,i}| \mathbb{E}(|U_{t,j-i}|\,|\,\mathscr{F}_t).$$

Let $G_1 = 0$ and $G_j = \sum_{k=0}^{j-2} L^k$ for $j \ge 2$. The bound $\mathbb{E}(|U_{t,1}|\,|\,\mathscr{F}_t) \le MVG_1$ holds since $U_{t,1} = 0$. If it holds for all horizons below j, then

$$\begin{aligned} \mathbb{E}(|U_{t,j}|\,|\,\mathscr{F}_t) &\le MV \left(1 + \sum_{i=1}^{p_j} |\phi_{t,j,i}| G_{j-i} \right) \\ &\le MV(1 + LG_{j-1}) = MVG_j, \end{aligned}$$

where $G_{j-i} \le G_{j-1}$ because $L \ge 0$. Strong induction therefore establishes the bound for all $1 \le j \le h$, and taking $j = h$ proves Equation (10). □

A4 Approximation error analysis for Propositions 1 and 2

Omitting the remainders in Propositions 1 and 2 yields linear approximations with errors $\widetilde{R}_{t,h}$ and $U_{t,h}$. Equation (8) controls the first through curvature and shorter-horizon conditional mean squared errors; Equation (10) additionally accounts for propagation through G_h, defined in Appendix A3. For fixed $L \ge 0$, this factor is bounded by $(1-L)^{-1}$ when $L < 1$, equals $h-1$ when $L = 1$, and grows geometrically with h when $L > 1$.

For the small-perturbation analysis, fix t, h, d with $h \ge 2$ and consider a family of processes indexed by $\delta > 0$, each satisfying the assumptions for Equation (7) in Proposition 1. Suppose that the following bounds hold almost surely for finite deterministic constants $M, K, \sigma \ge 0$ independent of δ:

1. The Hessian bounds in Proposition 1 hold with $M_{t,j} \le M$ for $2 \le j \le h$.

2. The gradients along the expansion segments satisfy

$$\sup_{u\in[0,1]} \left\| \nabla F_{t,j}\big(\boldsymbol{m}_{t,j} + u(\boldsymbol{z}_{t,j} - \boldsymbol{m}_{t,j})\big) \right\|_2 \le K, \qquad 2 \le j \le h.$$

3. The innovation conditional variances satisfy $\mathbb{E}(\varepsilon_{t+j}^2 | \mathscr{F}_{t+j-1}) \le \sigma^2\delta^2$ for $1 \le j \le h$.

Dependence on δ is suppressed in the notation.

By Cauchy–Schwarz, $\sum_{i=1}^{p_j} |\phi_{t,j,i}| \le \sqrt{d}K$, so $L = \sqrt{d}K$ is admissible.

Write $V_j = \mathbb{E}(e_{t+j|t}^2 | \mathscr{F}_t)$ for $1 \le j \le h$. Conditional mean optimality gives, for $j \ge 2$,

$$V_j \le \mathbb{E}\big((y_{t+j} - F_{t,j}(\boldsymbol{m}_{t,j}))^2 \,|\, \mathscr{F}_t\big)\,.$$

Since $F_{t,j}(\boldsymbol{z}_{t,j}) - F_{t,j}(\boldsymbol{m}_{t,j})$ is $\mathscr{F}_{t+j-1}$-measurable, its cross term with ε_{t+j} has conditional expectation zero. The gradient bound gives $|F_{t,j}(\boldsymbol{z}_{t,j}) - F_{t,j}(\boldsymbol{m}_{t,j})| \le K\|\boldsymbol{\Delta}_{t,j}\|_2$, using the increment defined in Appendix A2. Together with the innovation variance bound, this yields

$$V_1 \le \sigma^2\delta^2, \qquad V_j \le \sigma^2\delta^2 + K^2 \sum_{i=1}^{p_j} V_{j-i}, \qquad 2 \le j \le h.$$

Induction gives $V_j \le A_j\delta^2$, where $A_1 = \sigma^2$ and $A_j = \sigma^2 + K^2\sum_{i=1}^{p_j} A_{j-i}$. Taking $C = \max_{2\le j\le h}\sum_{i=1}^{p_j} A_{j-i}$, independent of δ, and substituting $V = C\delta^2$ into the remainder bounds yields, almost surely,

$$\mathbb{E}(|\widetilde{R}_{t,j}|\,|\,\mathscr{F}_t) \le MC\delta^2, \qquad 2 \le j \le h, \qquad \mathbb{E}(|U_{t,h}|\,|\,\mathscr{F}_t) \le MCG_h\delta^2.$$

Taking expectations shows that, under the stated conditions, both remainders have mean absolute size $O(\delta^2)$ and are negligible relative to δ. Specifically, $\mathbb{E}|\widetilde{R}_{t,j}/\delta| \to 0$ for $2 \le j \le h$ and $\mathbb{E}|U_{t,h}/\delta| \to 0$ as $\delta \downarrow 0$.

These results quantify linear approximation errors as the innovation scale decreases at fixed forecast horizons, rather than as the training-sample size increases. At a given perturbation scale, large curvature or propagation factors may make the bounds uninformative.

A5 Proof of Proposition 3

Proof. Let $E_T = \sum_{i=h+1}^{T}(\mathrm{err}_{i|i-h} - \alpha)$. The inequality in Equation (12) is equivalent to $|E_T| \leq cg(T-h)+h$. We first prove the upper bound by strong induction, with the lower bound following analogously.

For $T = h+1,\ldots,2h$, $|E_T| \leq T-h \leq h \leq cg(T-h)+h$, since g is nonnegative and $0 \leq \mathrm{err}_{i|i-h} \leq 1$. Thus both bounds hold over the initial range.

Now let $T \geq 2h+1$ and assume the upper bound holds at all earlier indices. The threshold evaluated at time T is

$$q_{T|T-h} = r_{T-h}(E_{T-h}).$$

We partition the argument into two cases. If $E_{T-h} \geq cg(T-2h)$, Equation (5) gives $q_{T|T-h} \geq b$, so $\mathrm{err}_{T|T-h} = 0$. By the induction hypothesis and monotonicity of g,

$$E_T = E_{T-1} - \alpha \leq cg(T-1-h)+h \leq cg(T-h)+h.$$

Otherwise, using $\mathrm{err}_{i|i-h} - \alpha \leq 1-\alpha$ over the last h observations gives

$$E_T = E_{T-h} + \sum_{i=T-h+1}^{T}(\mathrm{err}_{i|i-h} - \alpha) < cg(T-2h)+h(1-\alpha) \leq cg(T-h)+h.$$

This completes the induction for the upper bound. For the lower bound, the same induction applies: if $E_{T-h} \leq -cg(T-2h)$, saturation and $s_{T|T-h} > -b$ imply $\mathrm{err}_{T|T-h} = 1$, so

$$E_T = E_{T-1} + 1 - \alpha \geq -cg(T-1-h) - h \geq -cg(T-h) - h.$$

Otherwise, $\mathrm{err}_{i|i-h} - \alpha \geq -\alpha$ yields

$$E_T > -cg(T-2h) - h\alpha \geq -cg(T-h) - h.$$

The argument also applies when $b = \infty$ because every score is finite. Dividing by $T-h$ proves Equation (12).

Regarding the long-run conclusion, $g(T-h)/(T-h) \to 0$ by sublinearity, and $h/(T-h) \to 0$

for fixed h. Hence the empirical tail miscoverage rate converges to α. □

A6 Proof of Corollary 1

Proof. Unraveling the recursion gives

$$q_{t+h|t} = q_{2h|h} + \eta \sum_{i=h+1}^{t} (\mathrm{err}_{i|i-h} - \alpha).$$

Thus, take $r_t(x) = q_{2h|h} + \eta x$, $g \equiv 1$, and $c = (b + |q_{2h|h}|)/\eta$. If $x \geq c$, then $r_t(x) \geq b$, and if $x \leq -c$, then $r_t(x) \leq -b$, so Equation (5) holds. Applying Proposition 3 gives the stated bound and long-run calibration. □

A7 Proof of Corollary 2

Proof. Let $s^*_{t+h|t} = s_{t+h|t} - \hat{q}_{t+h|t}$ and $q^*_{t+h|t} = q_{t+h|t} - \hat{q}_{t+h|t}$. Since the scores and baseline values are finite, these transformations are well-defined even when the thresholds are infinite, and $s^*_{t+h|t} \in (-b, b)$ by assumption. Subtracting the same finite baseline preserves the tail miscoverage indicator: $\mathrm{err}_{t+h|t} = \mathbb{1}\{s^*_{t+h|t} > q^*_{t+h|t}\}$. Equation (13) therefore becomes

$$q^*_{t+h|t} = r_t \left(\sum_{i=h+1}^{t} \left(\mathbb{1}\{s^*_{i|i-h} > q^*_{i|i-h}\} - \alpha \right) \right).$$

The transformed recursion satisfies the assumptions of Proposition 3 with the same b, c, and g. Applying that proposition gives the stated finite-sample bound and long-run calibration for the unchanged indicators. □

A8 Proof of Corollary 3

Proof. Common saturation constants can be chosen by taking the maximum of the two constants and the pointwise maximum of the two admissible functions. Apply Corollary 2 separately to the lower- and upper-tail recursions, replacing its generic target α by α^- and α^+, respectively. With

$n = T - h$, this gives

$$\left| \frac{1}{n} \sum_{t=h+1}^{T} \mathrm{err}^{a}_{t|t-h} - \alpha^{a} \right| \leq \frac{c \cdot g(n) + h}{n}, \qquad a \in \{-, +\}.$$

The interval fails to cover y_t if and only if at least one tail constraint fails, including when the interval is empty. Thus,

$$\mathbb{1}\{y_t \notin \hat{\mathscr{C}}^{\alpha}_{t|t-h}\} = \mathrm{err}^{-}_{t|t-h} + \mathrm{err}^{+}_{t|t-h} - \mathrm{err}^{-}_{t|t-h}\mathrm{err}^{+}_{t|t-h}.$$

Dropping the nonnegative overlap term, summing, and using $\alpha^{-} + \alpha^{+} = \alpha$ yields

$$\frac{1}{n} \sum_{t=h+1}^{T} \mathbb{1}\{y_t \notin \hat{\mathscr{C}}^{\alpha}_{t|t-h}\} \leq \alpha + 2\frac{c \cdot g(n) + h}{n},$$

which is equivalent to Equation (14). Because g is sublinear and h is fixed, $\big(c \cdot g(n) + h\big)/n \to 0$, giving the long-run coverage lower bound.

For the absolute deviation bound, simultaneous violations of both tail bounds imply

$$q^{+}_{t|t-h} < y_t - \hat{y}_{t|t-h} < -q^{-}_{t|t-h},$$

so $\hat{y}_{t|t-h} - q^{-}_{t|t-h} > \hat{y}_{t|t-h} + q^{+}_{t|t-h}$, a crossing of the displayed endpoints. This comparison remains well-defined for infinite thresholds. Thus the average of the products of the two tail indicators is at most $\chi_{T,h}$. The union identity and the triangle inequality give

$$\begin{aligned} \left| \frac{1}{n} \sum_{t=h+1}^{T} \mathbb{1}\{y_t \in \hat{\mathscr{C}}^{\alpha}_{t|t-h}\} - (1-\alpha) \right| &\leq \sum_{a \in \{-,+\}} \left| \frac{1}{n} \sum_{t=h+1}^{T} \mathrm{err}^{a}_{t|t-h} - \alpha^{a} \right| + \frac{1}{n} \sum_{t=h+1}^{T} \mathrm{err}^{-}_{t|t-h}\mathrm{err}^{+}_{t|t-h} \\ &\leq 2\frac{c \cdot g(n) + h}{n} + \chi_{T,h}. \end{aligned}$$

This proves Equation (15) and exact long-run calibration when $\chi_{T,h} \to 0$.

For the iteration in Proposition 3, the same proof applies with zero baselines. □

A9 Two-sided coverage for constant-step quantile tracking

Corollary 4. *Fix $h \in [H]$ and targets $\alpha^{-}, \alpha^{+} > 0$ with $\alpha^{-} + \alpha^{+} = \alpha \in (0,1)$. Apply the iteration in Corollary 1 separately to the two signed tail scores and construct the interval in Equation (2).*

Suppose each tail $a \in \{-,+\}$ satisfies the assumptions of Corollary 1 with finite score bound b^a, step size $\eta^a > 0$, and finite initial threshold $q^a_{2h|h}$. Then, for every $T \geq h+1$,

$$\frac{1}{T-h}\sum_{t=h+1}^{T} \mathbb{1}\{y_t \in \hat{\mathscr{C}}^{\alpha}_{t|t-h}\} \geq 1-\alpha - \sum_{a\in\{-,+\}} \frac{b^a + |q^a_{2h|h}| + \eta^a h}{\eta^a (T-h)}. \tag{19}$$

The absolute deviation of the empirical coverage from $1-\alpha$ is bounded by the sum on the right plus the endpoint-crossing frequency $\chi_{T,h}$ defined in Corollary 3. Consequently, the limit inferior of the empirical coverage is at least $1-\alpha$, and the coverage converges to $1-\alpha$ if $\chi_{T,h} \to 0$.

Proof. With $n = T-h$, Corollary 1 gives

$$\left|\frac{1}{n}\sum_{t=h+1}^{T} \mathrm{err}^a_{t|t-h} - \alpha^a\right| \leq \frac{b^a + |q^a_{2h|h}| + \eta^a h}{\eta^a n}, \qquad a \in \{-,+\}.$$

Applying the union identity and endpoint-crossing argument in Appendix A8 gives Equation (19) and the absolute deviation bound. For fixed h, the two tail bounds tend to zero, giving the stated long-run conclusions. □

A10 Local coverage bounds

Fix $h \in [H]$ and a window $\{a+1,\ldots,a+m\}$, where $a \geq h$ and $m \geq 1$ are integers. Under the assumptions of Corollary 2, set $D_h = 0$ and $D_u = c \cdot g(u-h) + h$ for $u \geq h+1$. Differencing the cumulative deviations at $a+m$ and a and applying the triangle inequality yields

$$\left|\frac{1}{m}\sum_{t=a+1}^{a+m} (\mathrm{err}_{t|t-h} - \alpha)\right| \leq \frac{D_{a+m} + D_a}{m}. \tag{20}$$

Under the assumptions of Corollary 3, with common c and g for both tails and $\alpha = \alpha^- + \alpha^+$, the union bound gives

$$\frac{1}{m}\sum_{t=a+1}^{a+m} \mathbb{1}\{y_t \in \hat{\mathscr{C}}^{\alpha}_{t|t-h}\} \geq 1-\alpha - 2\frac{D_{a+m} + D_a}{m}. \tag{21}$$

If $(D_{a+m} + D_a)/m \to 0$ along a sequence of windows, empirical window coverage has limit inferior at least $1-\alpha$ and converges to $1-\alpha$ if the within-window endpoint-crossing frequency also vanishes, by Appendix A8. These bounds may be conservative and uninformative for fixed-length

windows.

A11 Sensitivity to uncertainty in future predictors

We assess the effect of unknown future predictors using

$$x_t = 0.7x_{t-1} + u_t, \qquad y_t = 0.8y_{t-1} - 0.5y_{t-2} + x_t + \varepsilon_t,$$

where $u_t \overset{\text{iid}}{\sim} N(0, \tau^2)$ and $\varepsilon_t \overset{\text{iid}}{\sim} N(0,1)$ are mutually independent, with $\tau \in \{0.25, 1\}$. Each of 100 replications retains 5000 observations after a 500-observation burn-in. At each forecast origin, an ARX(2) model with an intercept is estimated by least squares over a rolling window of 500 observations. We compare forecasts using observed future predictors (Known) with forecasts using $\hat{x}_{t+j|t} = 0.7^j x_t$ (Forecast). The autoregressive coefficient of x_t is treated as known to isolate future predictor uncertainty from its estimation error.

Both settings use the same simulated paths and the AcMCP settings in Section 5.1, but are calibrated separately. We evaluate horizons $h = 1, 2, 3$ at a target coverage of 90%. Local coverage MAE measures the average absolute deviation of 500-observation rolling coverage from 90%.

Table A1 shows that both settings maintain coverage close to 90%, with local coverage MAE below 0.3 percentage points. Interval widths in the Known setting are nearly unchanged across the two values of τ, whereas forecasting future predictors leads to wider intervals, with larger increases at higher τ and longer horizons. At $h = 3$, the increase is about 8% for $\tau = 0.25$ and 97% for $\tau = 1$. Thus, in this simulation, AcMCP accommodates predictor uncertainty primarily through wider intervals while maintaining empirical coverage.

Table A1: *AcMCP with observed (Known) and forecast (Forecast) future predictors. Entries are means over 100 paired simulation replications, with Monte Carlo standard errors in parentheses. pp denotes percentage points.*

	Coverage (%)		Mean interval width		Local coverage MAE (pp)	
	Known	Forecast	Known	Forecast	Known	Forecast
			$\tau = 0.25$			
$h = 1$	89.99 (0.003)	89.99 (0.004)	3.55 (0.005)	3.65 (0.005)	0.20 (0.001)	0.19 (0.001)
$h = 2$	89.99 (0.004)	89.99 (0.004)	4.72 (0.008)	5.00 (0.010)	0.24 (0.002)	0.24 (0.002)
$h = 3$	89.99 (0.004)	89.99 (0.005)	4.85 (0.009)	5.26 (0.011)	0.27 (0.002)	0.27 (0.002)
			$\tau = 1$			
$h = 1$	89.99 (0.003)	89.99 (0.003)	3.56 (0.005)	4.99 (0.007)	0.20 (0.001)	0.19 (0.001)
$h = 2$	89.99 (0.004)	89.99 (0.004)	4.72 (0.009)	8.10 (0.016)	0.24 (0.002)	0.24 (0.001)
$h = 3$	89.98 (0.005)	89.99 (0.005)	4.85 (0.009)	9.57 (0.021)	0.27 (0.002)	0.29 (0.002)